Boundary coherence and work fluctuations under quantum confinement control

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Coherent boundary interference can purify the retained part of an increasingly mixed Gibbs state, while its work statistics probe inequivalent spectral sectors. We establish this connection for a confined particle subject to a coherently controlled potential wall. A common boundary source governs the conditional state, while its positive spectral measure determines the leading population–coherence allocation of dissipated work and the Ramsey response. Resonant interference produces an asymptotically pure retained state with an explicitly paid success probability. For a sharp wall, a discrete near-elastic sector contributes a logarithm to absolute work that is absent from the quadratic-weight continuum limit. Its contribution makes the ratio of absolute to net work diverge on specified thermal paths. Smooth-wall thermodynamic limits, a finite smooth-device certificate with timing and readout errors, and a finite-count reconstruction scheme connect these results to operational resources.

I. INTRODUCTION

Coherent control and thermal averaging interrogate different aspects of a quantum boundary. Thermal occupation counts the states available in an extended well. A local change of the wall instead couples to the small part of each wave function that penetrates the controlled region. It is therefore possible for many orthogonal occupied states to have nearly parallel responses to the same local perturbation. What remains to be established is whether this alignment survives the complete Gibbs ensemble, how much a retained state costs, and which work statistics can be inferred from its interferometric signature.

The characteristic function of quantum work can be measured by a Ramsey interferometer, without carrying out the two projective energy measurements in each run [1–3]. Conditional purification through repeated ancillary measurements is also well established [4], including thermal-state filtering by Ramsey postselection [5] and engineered ancilla-mediated logical-state protocols [6]. These observations motivate the experiment considered here, but do not settle its continuum limit. In particular, a vanishing unnormalized error need not remain small after conditioning on a rare outcome. Nor does convergence of purity to one imply convergence of entropy in an unbounded Hilbert space. The retained state, its full success probability, and its spectral tails must be controlled together.

Another distinction concerns the information contained in work. The decomposition of irreversible entropy production into population and coherence contributions is established in quantum thermodynamics [7]. In an extended continuum system, the thermal scale can compete with the exponentially small Gibbs weight at a fixed final energy. The boundary contribution must then be followed through the moving crossover rather than expanded uniformly in the transition probability. Finite-dimensional high-temperature quench expansions [8] address a different limit: here the barrier is thermally high and the number of populated low modes diverges. We obtain an explicit capped spectral moment for this competition, including the thermally activated population above the initial barrier.

The same system also reveals a limitation of inferring singular work functionals from a regular interferometric limit. A quadratic work weight gives a finite limiting measure. An absolute-value weight is more sensitive to transitions between nearby levels. We calculate their accumulated contribution directly in the finite well. It is logarithmic in the number of thermally occupied modes and survives after the regular continuum contribution has been extracted. Thus fixed-time convergence of a Ramsey response does not justify interchanging the continuum limit with an unbounded observation-time integral. Local-quench work edges and scattering-controlled Ramsey responses already occur in critical and Fermi systems [9, 10, 12]. Scattering-induced logarithms are also central to orthogonality catastrophe [11]. The present calculation instead concerns a single-particle Gibbs ensemble, a finite wall-height ratio, and a gap-weighted thermal harmonic sum with an additive remainder. Its finite-amplitude quench and singular thermal limit distinguish it from weak-amplitude work response [13].

In an earlier work [17], we established finite-wall boundary-source laws and rare work costs at fixed inverse temperature in box units. Here that inverse temperature tends to zero, and the number of thermally occupied well modes diverges jointly with the barrier scale. The resulting coherence is leading order, the relevant thermal crossover is retained, and the nearby-level work sector requires a separate calculation. The physical instrument remains a pair of controlled propagators; no final motional-energy projection or commuting surrogate filter is inserted. The analysis yields four connected results. It identifies the common boundary spectral measure, determines its thermodynamic allocation, resolves the singular absolute-work law for a sharp wall, and gives finite operational statements for purification and reconstruction. The complete derivations below make explicit where a

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smooth-profile result applies and where the sharp-wall calculation uses its exact matching conditions.

II. A CONTROLLED BOUNDARY AND ITS PHYSICAL SCALES

Let a particle occupy the interval $0 < y < 2$, with Dirichlet boundary conditions. Initially and on the controlled branch its dimensionless Hamiltonians are

$$A_r = -r^2\partial_y^2 + F((y-1)/r), \qquad B_r = A_r + 3F((y-1)/r). \tag{1}$$

The sharp wall is $F(x) = \mathbf{1}_{(0,\infty)}(x)$. The smooth asymptotic family consists of a fixed nondecreasing smooth F with $0 \le F \le 1$, $F = 0$ on $x \le -d$ and $F = 1$ on $x \ge d$, for a fixed $d > 0$. Constants may depend on this profile. A separate finite example will use an error-function wall with noncompact, exponentially small tails.

The prepared state is the full Gibbs state

$$\rho_{r,s} = Z_1^{-1}\mathrm{e}^{-sA_r}, \qquad Z_h = \operatorname{Tr}\mathrm{e}^{-s[-r^2\partial_y^2 + hF((y-1)/r)]}. \tag{2}$$

All initial eigenstates are included. A pure ancillary two-level system coherently selects the two propagators. Its dark outcome retains the motional operator

$$\begin{aligned} K_t &= \frac{\mathrm{e}^{-itB_r} - \mathrm{e}^{-itA_r}}{2}, \\ p_{r,s}(t) &= \operatorname{Tr} K_t \rho_{r,s} K_t^*, \\ \sigma_t &= K_t \rho_{r,s} K_t^* / p_{r,s}(t). \end{aligned} \tag{3}$$

The complementary Kraus operator is the sum of the same propagators divided by two. Their squared operators add to the identity.

To restore units, let the physical coordinate be $L_0 y$ and put $E_0 = \hbar^2/(2mL_0^2)$. Then

$$H_{A,B} = \frac{E_0}{r^2}(A_r, B_r), \qquad \beta_{\mathrm{phys}} = \frac{sr^2}{E_0}, \qquad t_{\mathrm{phys}} = \frac{\hbar r^2}{E_0}t. \tag{4}$$

The barrier height grows as r^{-2}, while a fixed dimensionless duration shrinks in laboratory units. Their product is a bounded action; this is not a bounded-height or fixed-duration protocol.

The basic limit and two useful parameters are

$$r \downarrow 0, \quad s \to \infty, \quad r\sqrt{s} \to 0, \qquad N = \frac{1}{r\sqrt{s}}, \quad \Xi = \frac{s\mathrm{e}^{-s}}{r}. \tag{5}$$

N measures the growing thermal mode count. The input purity obeys $\operatorname{Tr}\rho_{r,s}^2 \sim \sqrt{2\pi}\, r\sqrt{s}$ and hence tends to zero. The parameter Ξ compares activated population in the raised region with the boundary contribution to a rare output. Some results below require Ξ bounded or convergent; the resonant sharp-wall purification result does not.

III. ONE SPECTRAL MEASURE FOR THE BOUNDARY RESPONSE

In the boundary coordinate $x = (y-1)/r$ the interval is $(-\ell, \ell)$, $\ell = 1/r$, and the kinetic term is $-\partial_x^2$. For a smooth plateau profile let f_h be the positive decaying zero-energy solution on $[-d, \infty)$,

$$-f_h'' + hFf_h = 0, \quad f_h(-d) = 1, \quad m_h = -f_h'(-d) > 0. \tag{6}$$

Set $g = Ff_1$, extended by zero to $x < -d$, and $B_\infty = -\partial_x^2 + 4F$ on the full line. Define

$$\nu_F(\mathrm{d}E) = \frac{9}{m_1^2}E^{-2}\langle g, \mathbf{1}_{\mathrm{d}E}(B_\infty)g\rangle, \qquad E > 0. \tag{7}$$

This is a positive spectral measure, not a probability distribution. Its mass is infinite at zero, while its first and second moments are finite. Write

$$\begin{aligned} D_F &= m_1^{-1} - m_4^{-1}, \\ M_F &= 3\dot{m}_1/m_1^2 - D_F = \int E\,\nu_F(\mathrm{d}E), \\ Q_F &= 9\,\|g\|^2/m_1^2 = \int E^2\,\nu_F(\mathrm{d}E), \end{aligned} \tag{8}$$

where a dot denotes differentiation with respect to wall height. The two response functions used throughout are

$$\begin{aligned} C_F(t) &= \int \sin^2(tE/2)\,\nu_F(\mathrm{d}E), \\ \Phi_F(c) &= \int \min(E, c)\,\nu_F(\mathrm{d}E). \end{aligned} \tag{9}$$

The local density at threshold is

$$\nu_F'(E) = \frac{D_F^2}{\pi}E^{-3/2}[1 + O_F(E)] \quad (E \downarrow 0). \tag{10}$$

It follows that $C_F(t)$ is finite at every fixed time and $0 < \Phi_F(c) \le M_F$ for $c > 0$, with strict upper inequality for $0 < c < 4$. The coefficient D_F is unchanged by moving the reference point $-d$ farther into the free plateau. It satisfies $D_F \ge 1/2$, strictly for a genuine smooth transition. The resulting ordered limits are

$$\Phi_F(1/a) \sim \frac{4D_F^2}{\pi\sqrt{a}}, \qquad C_F(t) \sim \frac{D_F^2}{\sqrt{2\pi}}\sqrt{|t|}, \tag{11}$$

as $a \to \infty$ or $|t| \to \infty$ after forming the continuum source. These are not joint growing-time statements for the finite device.

For the sharp wall $m_h = \sqrt{h}$, $D_F = 1/2$, $M_F = 1$, $Q_F = 9/2$, and we omit the subscript F. Matching the full-line Green function gives the explicit density

$$\nu'(E) = \begin{cases} \dfrac{9}{4\pi E^{3/2}(1+\sqrt{4-E})^2}, & 0 < E < 4, \\[2ex] \dfrac{9(1+\sqrt{E}\sqrt{E-4})}{\pi E^2(\sqrt{E}+\sqrt{E-4})(E-3)^2}, & E > 4. \end{cases} \tag{12}$$

There is no atom at four. The apparent change there is the opening of the right propagating channel, not an additional bound state.

IV. COHERENCE RETAINED BY BOUNDARY INTERFERENCE

For a low initial eigenstate u_E, the part in the controlled region is a scalar boundary value times a common analytic family: $Fu_E = b_E g_{E,\ell}$. Duhamel's identity therefore reads exactly

$$\begin{aligned} K_t u_E &= b_E \mathrm{e}^{-itE} V_{E,\ell}(t), \\ V_{E,\ell}(t) &= \frac{3}{2} h_t(B_\ell - E) g_{E,\ell}, \\ h_t(z) &= \frac{\mathrm{e}^{-itz} - 1}{z}. \end{aligned} \tag{13}$$

The value at zero is $h_t(0) = -it$. Low thermal energies approach zero, so these columns become parallel even though their input eigenstates are orthogonal. Their complete thermal covariance has leading term

$$K_t \rho_{r,s} K_t^* \big|_{\text{low input}} = \frac{r}{s m_1^2} |V_{0,\ell}(t)\rangle\langle V_{0,\ell}(t)| + o_1(r/s), \tag{14}$$

where o_1 is in trace norm. The source vector converges strongly to $V_t = (3/2) h_t(B_\infty) g$, with $\|V_t\|^2 / m_1^2 = C_F(t)$.

If $\Xi \to x < \infty$, the full, off-resonant probability satisfies

$$\frac{s}{r} p_{r,s}(t) \longrightarrow C_F(t) + x \sin^2(3t/2), \tag{15}$$

uniformly on each fixed compact time interval. The second term is activated volume, not another coherent source. For a fixed $t \neq 0$, put $b = x \sin^2(3t/2)$. Its positive covariance has bounded limiting trace and vanishing operator norm after scaling by s/r. Consequently

$$\begin{aligned} \lambda_1(\sigma_t) &\to \frac{C_F(t)}{C_F(t) + b}, \\ \lambda_2(\sigma_t) &\to 0, \\ \operatorname{Tr} \sigma_t^2 &\to \left[\frac{C_F(t)}{C_F(t) + b} \right]^2. \end{aligned} \tag{16}$$

When $b > 0$, the output entropy diverges: a nonzero weight is distributed among a growing number of small eigenvalues. When $b = 0$, purity alone does not decide entropy.

At the fundamental resonance $\tau = 2\pi/3$, the activated plateau acquires the same branch phase. For the sharp wall, cancellation leaves only boundary terms at every initial energy, including at the barrier threshold. In the whole regime (5),

$$p_{r,s}(\tau) \sim C(\tau) \frac{r}{s}, \qquad \|\sigma_\tau - \Pi_{r,\tau}\|_1 \to 0, \quad S(\sigma_\tau) \to 0, \tag{17}$$

and all fixed positive-order Rényi entropies tend to zero. $\Pi_{r,\tau}$ projects onto the actual finite-box zero-energy source. The entropy conclusion uses all-index spectral-tail bounds, given in Appendix E, rather than a finite-dimensional continuity inequality. The smooth plateau version has the same yield $C_F(\tau) r/s$ and entropy conclusion, with profile-dependent constants.

The average number of fresh preparations per retained state is $s/[C_F(\tau) r]$ to leading order. Each preparation includes a pure ancilla and a readout; the retained purity is conditional. The initial state becomes more mixed while the successful output becomes pure, but the success probability tends to zero.

A. Changing the controlled boundary changes the output rank

The source picture can be tested without changing the input ensemble. On $(0, 3)$ let

$$A_r = -r^2 \partial_y^2 + Q_L + Q_R, \quad Q_L = \mathbf{1}_{(0,1)}, \quad Q_R = \mathbf{1}_{(2,3)}. \tag{18}$$

Compare control of $3Q_R$ with control of $3(Q_L + Q_R)$, at the same resonance and with the same complete Gibbs preparation. The limits are

Controlled walls	Right only	Both
$p/(r/s)$	$C(\tau)$	$2C(\tau)$
Nonzero output weights	1	$1/2, 1/2$
Output purity	1	$1/2$
Output entropy	0	$\log 2$

Both initial parity sectors are present in each experiment. With one controlled wall they feed the same boundary vector; with both controlled walls they feed orthogonal final parity sectors of equal limiting weight. This is a change of the physical controlled Hamiltonian, not a parity postselection. The calculation in Appendix F keeps the factor of two in both the boundary amplitudes and the total preparation probability.

B. Repeated retention and the price of imperfect interference

For any fixed positive integer m, m ideal successive dark outcomes apply K_τ^m to the same particle. The limiting source is $K_{\infty,\tau}^{m-1} V_\tau$, and

$$p_m \sim C_{m,F} \frac{r}{s}, \qquad C_{m,F} = \frac{\left\| K_{\infty,\tau}^{m-1} V_\tau \right\|^2}{m_1^2} > 0. \tag{19}$$

The normalized state again has vanishing entropy. The positivity in this formula follows from injectivity of the actual first-resonance Kraus operator, not a presumed dominant eigenvector of a commuting filter. No interchange with $m \to \infty$ is needed.

Real visibility loss is described by the completely positive map

$$\begin{aligned} \mathcal{J}_\nu(X) &= \nu K X K^* + (1 - \nu) \mathcal{D}(X), \\ \mathcal{D}(X) &= \frac{U_B X U_B^* + U_A X U_A^*}{4}, \end{aligned} \tag{20}$$

with $0 \le \nu \le 1$ and $\mathrm{Tr}\,\mathcal{D}(X) = \mathrm{Tr}\,X/2$. For two independent stages, with $\nu_j \in [0,1]$ and $\epsilon_j = 1-\nu_j$, sufficient source-retention conditions are

$$\epsilon_1 + \epsilon_2 \to 0, \qquad \epsilon_1\epsilon_2 = o(r/s). \tag{21}$$

For vanishing entropy the second condition strengthens to $\epsilon_1\epsilon_2 \log N = o(r/s)$. Under Eq. (21), this strengthened product condition is also necessary. The all-false branch carries a mixture of four unitary images of the full Gibbs state and hence an entropy $\log N + O(1)$. Independent repetition suppresses that branch quadratically, but cannot make its preparation or entropy cost disappear.

V. WHERE DISSIPATED WORK BECOMES POPULATION OR COHERENCE

Let $\gamma_B = \mathrm{e}^{-sB_r}/Z_4$ and let Δ_B completely dephase in the final energy basis. With natural logarithms, set

$$\begin{aligned} I &= \mathsf{D}(\rho\|\gamma_B), \\ P &= \mathsf{D}(\Delta_B\rho\|\gamma_B), \\ J &= \mathsf{D}(\rho\|\Delta_B\rho), \qquad I = P + J. \end{aligned} \tag{22}$$

The identity is the established population–coherence decomposition [7]. Here it applies to the unselected state immediately after a sudden change of Hamiltonian, independently of whether a Ramsey outcome is retained. I equals the mean dissipated work multiplied by the physical inverse temperature.

Write $\Lambda = \log(1/r)$. On the paths

$$\frac{s}{\Lambda} \to a \in [1,\infty), \qquad \Xi \to x \in [0,\infty), \tag{23}$$

where both a and x are finite, the complete smooth-profile result is

$$\frac{(I,P,J)}{r} \longrightarrow \big(M_F + 3x,\ M_F - \Phi_F(1/a) + 3x,\ \Phi_F(1/a)\big). \tag{24}$$

Necessarily $x = 0$ if $a > 1$; for example $s = a\Lambda$ realizes this case. At $a = 1$, any $x > 0$ is realized by $s = \Lambda + \log\Lambda - \log x$, while $s = \Lambda + 2\log\Lambda$ realizes $x = 0$. The activated sector changes the total and population terms by $3x$, while the leading coherence coefficient is unchanged. Its final-energy weight concentrates at four; its work increment concentrates at three. These are different variables.

The cap $1/a$ has a direct origin. At a fixed final energy E, the source population is polynomially small in the boundary parameter, whereas the equilibrium weight is proportional to e^{-sE}. Their competition occurs at $E = 1/a$. Below it, population relative entropy is negligible at order r; above it, the contribution is $(E - 1/a)\nu_F(\mathrm{d}E)$. This interpretation alone would not control the moving crossover or the infinitely many final levels. The proof in Appendix B pays both with positive diagonal excess and an energy-weighted tail estimate.

For the sharp wall, $M_F = 1$ and the coherent fraction of the leading source dissipation is $\Phi(1/a)$. At $a = 1$,

$$0.4040654186 < \Phi(1) < 0.4040654188. \tag{25}$$

For a smooth profile, $D_F > 1/2$ increases the sequential large-a coefficient in Eq. (11). This compares that coefficient, not all finite-a curves. A useful profile-independent connection to the retained yield is

$$C_F(2\pi/3) \le a\Phi_F(1/a) \le aM_F. \tag{26}$$

The first inequality follows pointwise from $\sin^2(\pi E/3) \le \min(E,1) \le a\min(E,1/a)$.

Although $I = O(r)$ tends to zero, its dimensional work need not. Equation (4) gives $\langle W_{\mathrm{diss}}\rangle \sim E_0(M_F + 3x)/(rs)$. The dimensionless relative entropy is not a fixed-energy cooling efficiency. Conditional retained states and unselected work statistics answer different operational questions.

VI. A REGULAR WORK LIMIT AND A SINGULAR ABSOLUTE-WORK LAW

Let (E_n, u_n) and (λ_j, ϕ_j) be the complete initial and final eigenpairs of Eq. (1). The sudden-quench work law in barrier units is

$$\Pi_{r,s} = \frac{1}{Z_1}\sum_{n,j} \mathrm{e}^{-sE_n}|\langle\phi_j, u_n\rangle|^2\delta_{\lambda_j - E_n}. \tag{27}$$

It is the two-projective-measurement work distribution. In our retained state protocol those energy measurements are not made: the spectral formula is also an exact expansion of the two controlled propagators. In particular,

$$p_{r,s}(t) = \int \sin^2(t\omega/2)\,\Pi_{r,s}(\mathrm{d}\omega). \tag{28}$$

Under Eq. (5) with $\Xi \to x < \infty$, the finite positive measures $\zeta_{r,s}(\mathrm{d}\omega) = (s/r)\omega^2\Pi_{r,s}(\mathrm{d}\omega)$ converge against every bounded continuous test function:

$$\zeta_{r,s} \implies \omega^2\nu_F(\mathrm{d}\omega) + 9x\delta_3. \tag{29}$$

The first term is supported on positive work. The limiting second moment is $Q_F + 9x$. Dividing the sine-square test function in Eq. (28) by ω^2, with its continuous value at zero, proves Eq. (15). Absolute work, however, requires the singular quotient $1/|\omega|$. Equation (29) cannot be used for that quotient.

For the sharp wall, direct finite-spectrum analysis re-

solves it. For every fixed $c > 0$, on Eq. (5) with $\Xi = O(1)$,

$$\frac{s}{r}\mathbb{E}\min(|\omega|, c) = \Phi(c) + \frac{1}{2} + \frac{\log N}{\pi^{3/2}\sqrt{s}} + \min(3, c)\Xi + o(1), \tag{30}$$

$$\frac{s}{r}\mathbb{E}|\omega| = \frac{3}{2} + \frac{\log N}{\pi^{3/2}\sqrt{s}} + 3\Xi + o(1), \tag{31}$$

$$\frac{s}{r}\mathbb{E}\omega = \frac{3}{2} + 3\Xi + o(1),$$

$$\frac{s}{r}\mathbb{E}(-\omega)_+ = \frac{\log N}{2\pi^{3/2}\sqrt{s}} + o(1). \tag{32}$$

The remainders are additive even when the displayed logarithmic term diverges. The 1/2 term comes from paired low levels whose work increments shrink to zero. The logarithm comes from a harmonic sum of transitions between nearby but differently indexed low levels. Neither term is obtained by applying a singular test function to Eq. (29). Appendix C derives them before taking the continuum limit.

Physical work is $W = (E_0/r^2)\omega$. The following consequence is independent of the unit used for energy. If $s/\Lambda \to a > 1$, then $\Xi \to 0$ and

$$\frac{\mathbb{E}|W|}{\mathbb{E}W} \sim \frac{2\sqrt{s}}{3a\pi^{3/2}}, \qquad \frac{\mathbb{E}(-W)_+}{\mathbb{E}(W)_+} \longrightarrow 1. \tag{33}$$

Large positive and negative absolute components increasingly cancel in the positive net work. Negative individual two-measurement outcomes are compatible with the nonnegative potential increment $B - A = 3F$; the mean is $3\,\mathrm{Tr}\,\rho F \geq 0$. There is no violation of passivity or of the positivity of dissipated work.

The near-elastic sector is infrared *relative to the barrier energy*. It need not have small laboratory energy. For example, a typical adjacent spacing is of order $r/\sqrt{s}$ in ω, hence $E_0/(r\sqrt{s})$ in W, which grows in Eq. (5) at fixed E_0. Similarly, fixed t in Eq. (15) means a shrinking laboratory duration. Equations (31)– (33) avoid interpreting a choice of units as a new low-frequency laboratory effect.

VII. RECOVERING A COHERENT COEFFICIENT FROM ACTUAL COUNTS

For any positive measure on $(0, \infty)$ with finite first moment, cosine integration gives the positive reconstruction

$$\Phi_F(c) = \frac{4}{\pi}\int_0^\infty \frac{1 - \cos(ct)}{t^2} C_F(t)\,\mathrm{d}t. \tag{34}$$

Its kernel has total mass $2c$. Thus an all-time uniform additive error η in C_F changes the reconstructed coefficient by at most $2c\eta$. This stability concerns a fixed capped moment; identifying the whole measure by differentiating cap data is a different inverse problem. Finite-resolution work measurements and coherence inference from energy data have their own established frameworks [14, 15]. Here the target is the leading known-model continuum coefficient, not exact finite-state coherence tomography or a replacement for measurement backaction.

The finite work law has the same integral identity with $C_F(t)$ replaced by $p_{r,s}(t)$ and $\Phi_F(c)$ by $\mathbb{E}\min(|\omega|, c)$. Equations (30) and (15) show why the scaled finite integral and the continuum-source integral cannot be interchanged. An infrared-rejected functional gives a regular bridge: for fixed $0 < \eta < c$,

$$\frac{s}{r}\mathbb{E}\{\min(|\omega|, c) - \min(|\omega|, \eta)\} \longrightarrow \Phi_F(c) - \Phi_F(\eta) + x[\min(3, c) - \min(3, \eta)]. \tag{35}$$

Its quotient by ω^2 is bounded and continuous. The device limit is taken before $\eta \downarrow 0$. The corresponding time kernel is $4[\cos(\eta t) - \cos(ct)]/(\pi t^2)$ and need not be positive.

For a finite positive source quadrature, let t_i be the midpoints of a partition of $[0, T]$ and

$$w_i = \frac{4}{\pi}\int_{I_i} \frac{1 - \cos(ct)}{t^2}\,\mathrm{d}t, \qquad W = \sum_i w_i \leq 2c, \qquad \lambda = s/r. \tag{36}$$

The actual device quantity is $S = \lambda\sum_i w_i p_{r,s}(t_i)$. Choose node i with probability w_i/W, independently on each fresh preparation, and record whether the dark outcome occurs. For independent trials with the same node response probabilities, after n trials the pooled count is exactly binomial with parameter $\bar{p} = S/(\lambda W)$. The estimator is $\widehat{S} = \lambda W k/n$. If a deterministic full-device certificate gives $d_- \leq S - \Phi_F(c) \leq d_+$, any exact binomial confidence interval $[L_n(k), U_n(k)]$ gives

$$\Phi_F(c) \in [\lambda W L_n(k) - d_+,\ \lambda W U_n(k) - d_-]. \tag{37}$$

There is no subtraction of a fitted model mean from the raw counts.

A convenient sufficient preparation budget, when $S \leq D$ is certified, is

$$n \geq \frac{2\lambda W \log(2/\alpha)}{\epsilon^2}(D + \epsilon/3). \tag{38}$$

It guarantees $|\widehat{S} - S| \leq \epsilon$ with probability at least $1 - \alpha$. The rare-event cost is proportional to s/r, not its square. It nevertheless remains large. For the sharp wall, $r = 1/200$, $s = 128$, $c = 1$, $T = 32$, and 256 midpoint nodes of spacing 1/8, certified continuum enclosures give two distinct designs:

Systematic accounting	Preparations	Total error
Signed, fixed device	62 217 817	< 0.050480
Separate absolute errors	5 999 715 692	< 0.099933

Both have confidence at least 95%. The second is less than $\Phi(1)/4$. The first uses opposing source-grid and finite-device biases at the certified point; the second pays

their absolute sizes separately. These counts include unsuccessful trials, fresh Gibbs states, pure ancillas, and readouts. They are sufficient mathematical budgets, not observed data, optimal lower bounds, or a claim of practical experimental advantage.

Resonance-only sampling does not generally determine the capped coefficient: $\sin^2(n\tau(E+3)/2) = \sin^2(n\tau E/2)$. For example, adding respectively $\beta\delta_{7/2}$ and $\beta(\delta_{1/2} + \delta_{13/2})/2$ to a common positive source preserves the first moment and every resonant sample but changes $\Phi(1)$ by $\beta/4$. This is an obstruction in the positive-measure class, not a claim to have constructed two actual wall profiles with identical resonant data. Off-resonant measurements and their activated background must be treated explicitly.

VIII. A FINITE SMOOTH DEVICE WITH UNRECORDED TIMING ERROR

The asymptotic results do not specify an experimental onset. A separate validated calculation includes the actual analytic wall

$$\begin{aligned} F_r(y) &= \tfrac{1}{2}\{1 + \mathrm{erf}[(y-1)/0.025]\}, \\ r &= 0.05, \quad s = 8, \\ A &= -r^2\partial_y^2 + F_r, \quad B = A + 3F_r. \end{aligned} \tag{39}$$

Its 10–90 percent transition width is $0.9061938024\ldots$ times the penetration length rL_0; it is not a vanishingly small perturbation of a sharp wall. The accepted output is averaged over any input-independent, unrecorded duration distribution supported on $2\pi/3 \pm 0.001$. Visibility is $\nu = 0.9999$ in Eq. (20). Readout independently has false-positive probability 10^{-5} conditional on a bright outcome, and false-negative probability 0.01 conditional on a dark outcome. For precisely this model, outward-rounded enclosures give

$$\begin{aligned} \mathrm{Tr}\,\rho^2 &< 0.386060, \\ \mathrm{Tr}\,\sigma^2_{\mathrm{recorded}} &> 0.9139989, \\ 0.00257508 < p_{\mathrm{recorded}} &< 0.00261934. \end{aligned} \tag{40}$$

The mean number of fresh preparations per recorded output is below 389. This expectation is not a high-probability guarantee or a tomography cost. For independent attempts, a lower bound p_- gives at least one success with probability $1-\alpha$ after $n \geq \log\alpha/\log(1-p_-)$ attempts.

The timing estimate is in the laboratory Schrödinger frame. For the actual Gibbs moments,

$$\begin{aligned} \|\partial_t K_t\sqrt{\rho}\|_2 &\leq \sqrt{\mathrm{Tr}\,\rho A^2} + \frac{3}{2}\sqrt{\mathrm{Tr}\,\rho F_r^2} \\ &= L < 0.220728. \end{aligned} \tag{41}$$

A purification retaining a classical duration register bounds the unnormalized amplitude error by $L\Delta t$; normalization by the rare success probability is then paid explicitly. A fixed-time interaction-picture bound alone would not establish this statement about a mixture of unrecorded durations.

The certificate uses validated ODE integration for the smooth transition, certified eigenvalue labels, complete Gibbs tails, and complete final spectral tails. Computational prefixes are not physical energy cutoffs. Appendix H gives the enclosure construction and its noise propagation; the public verification scripts reproduce the reported inequalities.

Physical resolution, temperature, and duration cannot be chosen independently. If the 10–90 edge width is $W = a_{\mathrm{erf}} r L_0$, where $a_{\mathrm{erf}} = 0.9061938024\ldots$, then

$$\begin{aligned} V &= \frac{\hbar^2 a_{\mathrm{erf}}^2}{2mW^2}, \quad k_B T = V/s, \\ t_{\mathrm{phys}} &= \tau\hbar/V, \quad k_B T t_{\mathrm{phys}}/\hbar = \tau/s. \end{aligned} \tag{42}$$

For ^{87}Rb and a 400-nm width, this example corresponds to approximately 1.79 nK and 1.12 ms; the timing half-width is about $0.533\,\mu$s. This conversion fixes a demanding target, not an apparatus demonstration. Finite outer walls, transverse excitations, state-dependent losses, preparation and reset overhead, and interactions require their own experimental error budgets.

IX. DISCUSSION

The common source measure organizes three physically different questions. A coherent boundary experiment retains a nearly pure vector because low-energy columns align. Unselected thermodynamics splits its leading source dissipation at a Gibbs crossover, producing a capped moment. A quadratic work weighting has a regular continuum limit, whereas the less regular absolute-value functional retains the discrete nearby-level sector. These statements are compatible precisely because they do not exchange conditioning, thermal limits, and singular spectral testing.

The finite smooth example makes conditional purification a quantitative prediction for a specified channel. The count construction addresses a different question: how many complete preparations suffice to infer a capped source coefficient. Its large cost is part of the result. Neither larger conditional purity nor a finite asymptotic coefficient alone establishes a measurement advantage.

Several experimentally meaningful comparisons remain possible within this framework: changing which wall is controlled at fixed preparation, measuring the dependence on visibility and timing spread, and comparing regular quadratic work statistics with the near-elastic contribution to absolute work. For smooth profiles the source and thermodynamic laws are proved directly; the sharp-wall logarithmic absolute-work coefficient is a separate exact matching result. Extending that coefficient to a prescribed smooth family would require a uniform two-energy scattering expansion, not merely its zero-energy

source density. This distinction keeps the present predictions tied to the Hamiltonians for which they have been established.

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DATA AVAILABILITY

Numerical results reported here are reproducible from the verification scripts and certificate data in the repository - Repository - github.com/arun-chandru/coherent_work.

Appendix A: Boundary source and regular work limit

We give the estimates used by the thermodynamic and work results, including the complete thermal sectors. Throughout this appendix $\ell = 1/r$, $b = \ell - d$, and $\mathcal{L} = \log(1/r)$; the latter must be distinguished from the boundary-coordinate length. Constants may depend on the fixed profile and on fixed energy cutoffs. For the sharp wall take $d = 0$ and use its elementary sine and hyperbolic-sine solutions. Write $A_{h,\ell} = -\partial_x^2 + hF$ with Dirichlet conditions on $(-\ell, \ell)$, and abbreviate $A_\ell = A_{1,\ell}$ and $B_\ell = A_{4,\ell}$.

1. Finite-box normalization and boundary amplitudes

Dirichlet and Neumann splitting at $-d, d$ gives, uniformly for $1 \le h \le 4$,

$$N_{h,\ell}(E) = \frac{b}{\pi}\big(\sqrt{E} + \sqrt{(E-h)_+}\big) + O_d(1+\sqrt{E}). \quad \text{(A1)}$$

The error is bounded by the free count on the fixed middle interval and the endpoint integer errors. The same bound holds for strict counts. Stieltjes integration therefore yields

$$\begin{aligned} Z_h &= \frac{b(1+\mathrm{e}^{-hs})}{2\sqrt{\pi}\sqrt{s}} + O_d(1+s^{-1/2}),\\ T_h &:= \operatorname{Tr} \mathbf{1}_{[h,\infty)}(A_{h,\ell})\mathrm{e}^{-sA_{h,\ell}}\\ &= \frac{b\mathrm{e}^{-hs}}{2\sqrt{\pi}\sqrt{s}}[1 + O_d(s^{-1/2} + \sqrt{s}/b)]. \end{aligned} \quad \text{(A2)}$$

For the second expression integrate the increment from $h-$: the right square root gives the displayed term, the left increment costs $O(b\mathrm{e}^{-hs}/s)$, and the counting error costs $O_d(\mathrm{e}^{-hs})$. Thus possible threshold eigenvalues are included. The same argument gives, for every fixed $e_0 > 0$,

$$Z_1^{-1} \operatorname{Tr} \mathbf{1}_{[e_0,\infty)}(A_\ell)\mathrm{e}^{-sA_\ell} \le C_{F,e_0}\mathrm{e}^{-se_0}. \quad \text{(A3)}$$

In the basic regime, $Z_h \sim (2\sqrt{\pi}r\sqrt{s})^{-1}$; applying this at s and $2s$ proves the input-purity formula in the main text.

The right Dirichlet operator $H_{h,D}$ on $(-d, \ell)$ has a uniform gap $\gamma_d = 1/\max(1, 2d^2)$: for $u(-d) = 0$, $\int_{-d}^{d} |u|^2 \le 2d^2 \int_{-d}^{d} |u'|^2$, while $hF \ge 1$ on (d, ℓ). Let $f_{h,E,\ell}$ equal one at $-d$, vanish at ℓ, and solve the right equation; set $m_{h,\ell}(E) = -f'_{h,E,\ell}(-d)$. Below a fixed small energy,

$$\begin{aligned} f_{h,E,\ell} &= (I - EH_{h,D}^{-1})^{-1} f_{h,0,\ell},\\ \partial_E m_{h,\ell}(E) &= -\left\| f_{h,E,\ell} \right\|^2. \end{aligned} \quad \text{(A4)}$$

The second identity follows by differentiating the equation and integrating its Wronskian. The zero-energy Dirichlet principle bounds $m_{h,\ell}(0) \ge (2d+1)^{-1}$: discard the transition potential and minimize $|1-u(d)|^2/(2d) + |u(d)|^2$. Zero-energy solutions have bounded norm, and the terminal constant plateau makes them and their low-energy derivatives converge exponentially as $\ell \to \infty$. Matching to the left sine, then integrating the normalization, gives

$$\begin{aligned} kb + \arctan\frac{k}{m_{h,\ell}(k^2)} &= n\pi,\\ |u_E(-d)|^2 &= \frac{2E}{b[E + m_{h,\ell}(E)^2] + m_{h,\ell}(E)}\\ &\qquad -2E\partial_E m_{h,\ell}(E)\\ &= \frac{2E}{bm_h^2}[1 + O_F(E + b^{-1} + \mathrm{e}^{-c_F b})]. \end{aligned} \quad \text{(A5)}$$

The phase count has bounded discrepancy from bk/π. For fixed nonnegative integers j, Gaussian Stieltjes integration implies

$$\sum_{E_n<\delta} \mathrm{e}^{-sE_n}|u_n(-d)|^2 E_n^j = \frac{\Gamma(j+3/2)}{\pi m_h^2} s^{-j-3/2}[1+o(1)], \quad \text{(A6)}$$

where $\delta > 0$ is fixed and sufficiently small. Relative errors are bounded by a constant times $s^{-1} + \sqrt{s}/b + \mathrm{e}^{-\delta s/2} + \mathrm{e}^{-c_F b}$, uniformly in h.

We shall also use the all-mode subthreshold estimates

$$\begin{aligned} \sum_{E_n<h} \left\| F^{1/2} u_n \right\|^2 &\le C_F,\\ \sum_{E_n<h} \mathrm{e}^{-sE_n} \left\| F u_n \right\|^2 &= O_F(s^{-3/2}). \end{aligned} \quad \text{(A7)}$$

For low energies these follow from Eq. (A5). For $\delta \le E < h$, left sine normalization is $O(b^{-1/2})$, and a

bounded transfer matrix across the fixed transition gives the same bound at d. The right-plateau mass is at most $C_F \min\{1, [b\sqrt{h-E}]^{-1}\}$. Integrate this increasing bounded weight against $N(E) = b\sqrt{E}/\pi + O_F(1)$. The principal integral is bounded by $\int_0^h [E(h-E)]^{-1/2} \mathrm{d}E = \pi$; the discrepancy is bounded by the total variation and endpoint value. The middle interval costs a constant by $\sum_{E_n \le 4} |u_n(x)|^2 \le C_F$. To prove this local bound apply the one-dimensional evaluation inequality to $(A_{h,\ell}+1)^{-1}$ and use its spectral expansion. This argument includes modes arbitrarily close to h. Multiplying the low sector by the heat weight and using Eq. (A6), then charging the rest exponentially, proves the second assertion, also with F in place of F^2.

2. Activated mass away from the controlled plateau

When $b/s \to \infty$, the following local estimate suffices:

$$\begin{aligned} U_h &:= \mathrm{Tr}(1-F)\mathbf{1}_{[h,\infty)}(A_{h,\ell})\mathrm{e}^{-sA_{h,\ell}} \\ &\le C_F \mathrm{e}^{-hs}(b/s+1) + C_F b \mathrm{e}^{-(h+1)s}/\sqrt{s}, \\ U_h/T_h &= O_F(s^{-1/2} + \sqrt{s}/b + \mathrm{e}^{-s}) = o(1). \end{aligned} \tag{A8}$$

Here is a local resolvent proof which avoids eigenfunction localization. On the free left plateau put $t = x + \ell \in (0, b)$, $z = k^2$, $k = p + iq$, $p, q > 0$. The right solution, normalized at $-d$, has logarithmic derivative $m(z) = \mathcal{E} - z\mathcal{N}$, where $\mathcal{E}, \mathcal{N}$ are its nonnegative energy and norm integrals. Writing $m = u - iv$ gives $qu + pv = q(\mathcal{E} + |k|^2\mathcal{N}) \ge 0$, hence $R(z) = [m(z) + ik]/[ik - m(z)]$ obeys $|R(z)| \le 1$. The diagonal Green function is exactly

$$G_z(t,t) = \frac{i}{2k} \frac{1 - \mathrm{e}^{2ikt} + R\mathrm{e}^{2ik(b-t)} - R\mathrm{e}^{2ikb}}{1 + R\mathrm{e}^{2ikb}}. \tag{A9}$$

For real parts in a fixed interval above zero and imaginary part $1/s$, one has $q \asymp 1/s$. The denominator stays away from zero when $b/s \to \infty$, so $|G_z(t,t)| \le C$. The spectral inequality $\mathbf{1}_{[v-\eta, v+\eta]}(A) \le 2\eta \,\mathrm{Im}(A - v - i\eta)^{-1}$ then gives a left-local count $Cb\eta$. Sum cells of width $1/s$ in $[h, h+1]$ with their heat weights to obtain $Cb\mathrm{e}^{-hs}/s$. The fixed middle interval costs $C_F \mathrm{e}^{-hs}$ by the preceding local spectral bound; Eq. (A3) pays every energy above $h+1$. This proves Eq. (A8).

For the sharp wall a frequency-bin bound removes the condition $b/s \to \infty$. On the physical left interval every mode above one is $a_n \sin(\omega_n y)$, $\omega_n = \sqrt{E_n}/r$. For sufficiently small r, the sine norm is at least $2/3$, and its frequency derivative has norm at most $2/3$. The normalized sine at the start of a frequency bin of width $1/4$ thus overlaps every sine in that bin by at least $1/2$. Bessel's inequality in the original full eigenbasis gives $\sum_{n \text{ in bin}} |a_n|^2 \le 4$. Counting bins yields

$$\begin{aligned} \sum_{1 \le E_n \le E} \|(1-F)u_n\|^2 &\le 4 + 8(E-1)/r, \\ T_- := \sum_{E_n \ge 1} \mathrm{e}^{-sE_n} \|(1-F)u_n\|^2 &\le \mathrm{e}^{-s}(4 + 8/(rs)). \end{aligned} \tag{A10}$$

The last step integrates the positive local spectral count, including a possible eigenvalue at one.

3. Source moments and threshold law

Zero-energy Wronskians give

$$\dot{m}_h = \int F f_h^2 \mathrm{d}x, \qquad 3\int F f_1 f_4 \mathrm{d}x = m_4 - m_1. \tag{A11}$$

Let $w = [f_1 - (m_1/m_4)f_4]/3$ on the right and extend it constantly to the left. Its derivative matches zero and $B_\infty w = g$ distributionally. Although the constant left tail generally prevents $w \in L^2$, its homogeneous energy is finite. Slow left cutoffs add arbitrarily little energy, while $|\int F f_1 v|^2 \le (\int F f_1^2)(\int F|v|^2)$ makes the source functional continuous in the energy norm. The form inverse, equivalently the monotone negative-resolvent limit, gives, with $\mu_g(\mathrm{d}E) = \langle g, \mathbf{1}_{\mathrm{d}E}(B_\infty) g\rangle$,

$$\int E^{-1} \mu_g(\mathrm{d}E) = \langle g, w\rangle = \frac{\dot{m}_1}{3} - \frac{m_1}{3m_4} \int F f_1 f_4 \mathrm{d}x. \tag{A12}$$

Multiplication by $9/m_1^2$ proves the first-moment identity in Eq. (8); the second is immediate from the source definition. Positivity is strict because $g \ne 0$.

The full-line operator has no eigenvalues: nonnegativity excludes negative ones and the free left equation excludes an L^2 solution at every nonnegative energy. For $0 < E < 4$, let h_E be the decaying right solution with $h_E(d) = 1$, $h_E'(d) = -\sqrt{4-E}$. Gluing to the outgoing free left solution gives

$$\begin{aligned} \mu_g'(E) &= \frac{\sqrt{E}}{\pi} \frac{|\int_{-d}^{\infty} g h_E \mathrm{d}x|^2}{h_E'(-d)^2 + E h_E(-d)^2}, \\ \mu_g(\mathrm{d}E) &= \langle g, \mathbf{1}_{\mathrm{d}E}(B_\infty) g\rangle. \end{aligned} \tag{A13}$$

One first derives this using a boundary-normalized right solution; the displayed form removes apparent poles at zeros of $h_E(-d)$. The denominator never vanishes. The numerator is real analytic and nonzero at zero by Eq. (A11), so each nonempty subinterval of $(0, 4)$ has positive source measure. Expansion at zero proves Eq. (10).

For the sharp wall put $f = \mathrm{e}^{-x}\mathbf{1}_{x>0}$, $\alpha = \sqrt{-z}$, $k = \sqrt{4-z}$ with positive real parts. Direct solution and derivative matching of $(B_\infty - z)v = f$ gives

$$\langle f, (B_\infty - z)^{-1} f\rangle = \frac{k + 2 + \alpha}{2(k+1)^2(k+\alpha)}. \tag{A14}$$

Upper-half-plane boundary values and Stieltjes inversion give Eq. (12), including the continuous value at four and the absence of an atom there. The value at $z = 0-$ is $1/9$, independently proving $\int E\nu(\mathrm{d}E) = 1$. Compact-interval inversion followed by exhaustion includes the entire high-energy tail.

Reference-point invariance follows because moving $-d$ left by a_0 sends m_h^{-1} to $m_h^{-1} + a_0$ and leaves both g/m_1 and $\dot{m}_h/m_h^2$ unchanged. To prove $D_F \geq 1/2$, put $u = f_h/m_h$, $v = -u'$. Then $v(-d) = 1$, $v \geq 0$, $v' = -hFu$, and $(v^2 - hu^2)' = 2h(1-F)uv \geq 0$, with terminal value zero. Thus $u \geq v/\sqrt{h}$ and

$$\frac{\dot{m}_h}{m_h^2} = -\frac{1}{h}\int uv'\mathrm{d}x \geq -\frac{1}{h^{3/2}}\int vv'\mathrm{d}x = \frac{1}{2h^{3/2}}. \quad \text{(A15)}$$

Integration from one to four proves the bound, strictly when a smooth transition has $0 < F < 1$ on an interval. Splitting the cap at $E = 1/a$ proves the first ordered limit in Eq. (11). For the second subtract $(D_F^2/\pi)E^{-3/2}\mathrm{d}E$, leaving a finite signed measure, and use $\int_0^\infty \sin^2(u/2)u^{-3/2}\mathrm{d}u = \sqrt{\pi/2}$.

4. The regular measure and fixed-time state limits

For a bounded continuous test f, the exact identity $(\lambda_j - E_n)\langle\phi_j, u_n\rangle = 3\langle\phi_j, Fu_n\rangle$ gives

$$\int f\mathrm{d}\zeta_{r,s} = 9\frac{s}{r}\sum_n \rho_n\langle Fu_n, f(B_\ell - E_n)Fu_n\rangle. \quad \text{(A16)}$$

At a zero gap the numerator vanishes as well. In the low sector write $Fu_n = b_n g_{E_n,\ell}$, $b_n = u_n(-d)$. Equation (A4) gives $\|g_{E,\ell} - g_{0,\ell}\| \leq C_F E$, and $g_{0,\ell} \to g$ after zero extension. Increasing Dirichlet intervals converge in strong resolvent sense. The associated vector spectral measures are tight because their masses converge and their compactly supported continuous tests converge. On a compact interval f is uniformly continuous, so the energy shift also converges, yielding

$$\langle g_{E,\ell}, f(B_\ell - E)g_{E,\ell}\rangle \longrightarrow \langle g, f(B_\infty)g\rangle \quad (E \to 0,\ \ell \to \infty). \quad \text{(A17)}$$

This argument does not assume global uniform continuity of f. Equations (A2) and (A6) give $(s/r)\sum_{E_n<\delta}\rho_n|b_n|^2 \to m_1^{-2}$, with vanishing weighted mass above any fixed positive low energy. Thus this sector in Eq. (A16) converges to $(9/m_1^2)\langle g, f(B_\infty)g\rangle$. The rest of the subthreshold sector is at most $C_F\|f\|_\infty s^{3/2}\mathrm{e}^{-s\delta}$ by Eq. (A7).

The complete activated sector is $O(\Xi)$ and hence vanishes if $x = 0$. If $x > 0$, then $\ell \sim x\mathrm{e}^s/s$, so Eq. (A8) applies. The exact shifted identity

$$(B_\ell - E_n - 3)u_n = -3(1-F)u_n \quad \text{(A18)}$$

implies concentration of its averaged work measure at three: the fraction outside $|\omega - 3| < \eta$ is bounded by $9\eta^{-2}$ times the averaged $\|(1-F)u_n\|^2 = o(1)$. Replacing Fu_n by u_n in Eq. (A16) costs at most $2\|f\|_\infty\|(1-F)u_n\|$ and vanishes by Cauchy–Schwarz. Since $(s/r)T_1/Z_1 \to x$, the result is $9xf(3)$. This proves Eq. (29), including its total mass. The same low calculation with F instead of F^2 gives

$$\frac{s}{r}\mathbb{E}\omega \to \frac{3\dot{m}_1}{m_1^2} + 3x = M_F + D_F + 3x. \quad \text{(A19)}$$

The continuous test $\sin^2(t\omega/2)/\omega^2$, valued $t^2/4$ at zero, proves Eq. (15). Uniformity on fixed bounded time intervals follows from $(s/r)|p'_{r,s}(t)| \leq (|t|/2)\zeta_{r,s}(\mathbb{R})$ and a finite time net. For the state version use Eq. (13): the same vector convergence and low moment give trace-norm convergence of the scaled low covariance to $m_1^{-2}|V_t\rangle\langle V_t|$. The collar vanishes by $\|K_t u_n\| \leq (3/2)|t|\,\|Fu_n\|$. The scaled activated covariance has trace tending to $b = x\sin^2(3t/2)$ and operator norm at most $\Xi/Z_1 \to 0$, since K_t is a contraction. Its squared Hilbert–Schmidt norm and its cross term with the source vanish. This proves Eq. (16). The source norm is positive for $t \neq 0$ by Eq. (A13). When $b > 0$, $S(\sigma_t) \geq -(1-\lambda_1(\sigma_t))\log\lambda_2(\sigma_t) \to \infty$. No entropy continuity is inferred when $b = 0$.

Appendix B: Complete static population and coherence allocation

We prove Eq. (24) for finite fixed $a \geq 1$ and finite $x \geq 0$ in Eq. (23). These paths have $b/s \to \infty$. All estimates below concern the full unselected sudden-quench state.

1. Total entropy and positive diagonal excess

The low thermal moments, Eq. (A11), and activated local estimate give, uniformly for $1 \leq h \leq 4$,

$$q_h := Z_h^{-1}\operatorname{Tr} F\mathrm{e}^{-sA_{h,\ell}} = \frac{r}{s}\frac{\dot{m}_h}{m_h^2}[1 + o(1)] + \mathrm{e}^{-hs}[1 + o(1)]. \quad \text{(B1)}$$

Each error is relative to its own positive term. Exact Duhamel trace differentiation, with $\kappa = Z_4/Z_1$, then gives

$$\begin{aligned} \log\kappa &= -s\int_1^4 q_h\mathrm{d}h = -rD_F + o(r), \\ 1 - \kappa &= rD_F + o(r), \\ I &= 3sq_1 + \log\kappa = rM_F + 3s\mathrm{e}^{-s} + o(r). \end{aligned} \quad \text{(B2)}$$

The integrated activated term is $O(\mathrm{e}^{-s}) = o(r)$. Entropies are finite at finite parameters: $\operatorname{Tr} B\rho < \infty$ follows from the bounded perturbation and the initial Gibbs mean energy; a Gibbs variational bound gives finite final diagonal entropy. Thus $I = P + J$ is an equality between finite quantities.

Write $p_j = \langle \phi_j, \rho \phi_j \rangle$, $q_j = Z_4^{-1} \mathrm{e}^{-s\lambda_j}$, $t_j = p_j/q_j$, and $d_j = p_j - \kappa q_j$. Scalar Jensen in the spectral measure of A for ϕ_j gives

$$d_j \geq 0, \quad t_j \geq \kappa, \quad \sum_j d_j = 1 - \kappa = O(r), \\ \sum_j |p_j - q_j| \leq 2(1-\kappa) = O(r). \tag{B3}$$

Indeed $\langle A - \lambda_j \rangle_j = -3\langle F \rangle_j$ and $\langle (A-\lambda_j)^2 \rangle_j = 9\langle F^2 \rangle_j$ exactly. No operator-monotonicity assertion for exponentials is used. Put $G(t) = t \log t - t + 1$. Then $P = \sum_j q_j G(t_j)$ has nonnegative summands and $G(t) \leq (t-1)\log t$ for $t \geq 1$. The whole sector $\kappa \leq t_j < 1$ costs $O((1-\kappa)^2) = O(r^2)$.

2. Low final energies and the moving crossover

For every fixed $c < 4$, $\sup_{\lambda_j \leq c} \langle F \rangle_j \leq C_{F,c} r$. At small energy this is Eq. (A5); above a small cutoff left sine normalization, compact transfer, and uniform right-plateau decay prove it. Taylor's scalar exponential bound and the two exact moments therefore imply, for $c < 1/a$,

$$\kappa \leq t_j \leq \kappa[1 + C_{F,c} r(s + s^2 \mathrm{e}^{sc})] = 1 + o(1) \quad (\lambda_j \leq c). \tag{B4}$$

Since $G(t) = o(1)|t-1|$ uniformly on this neighborhood of one, Eq. (B3) makes the complete low sea contribute $o(r)$.

Put $t_0 = 1/a$ and $\mathcal{K} = [3t_0/4, 5t_0/4] \subset (0,4)$. We require the pointwise bound

$$0 \leq d_j \leq C_{F,a} r^2/s \quad (\lambda_j \in \mathcal{K}). \tag{B5}$$

The nonnegative scalar remainder $R_{s,\lambda}(E) = \mathrm{e}^{-sE} - \mathrm{e}^{-s\lambda} + s\mathrm{e}^{-s\lambda}(E-\lambda)$ gives the exact identity

$$\langle \mathrm{e}^{-sA} \rangle_j - \mathrm{e}^{-s\lambda_j} = 3s\mathrm{e}^{-s\lambda_j} \langle F \rangle_j \\ + \sum_n R_{s,\lambda_j}(E_n) |\langle \phi_j, u_n \rangle|^2. \tag{B6}$$

Choose a fixed low cutoff δ below $\mathcal{K}$. For $E < \delta$, $R \leq \mathrm{e}^{-sE}$ and

$$(\lambda_j - E)\langle \phi_j, u_E \rangle = 3u_E(-d)\langle \phi_j, F f_{1,E,\ell} \rangle. \tag{B7}$$

The denominator is separated from zero; compact transfer and plateau decay bound the last inner product squared by Cr. Equation (A6) therefore pays this entire low sum by $Crs^{-3/2}$. For $E \geq \delta$, Taylor's theorem gives $R \leq (s^2/2)\mathrm{e}^{-s\delta}(E-\lambda_j)^2$; the exact second moment pays all remaining initial energies by $Crs^2\mathrm{e}^{-s\delta}$. The linear term is smaller. Division by $Z_1 \asymp 1/(r\sqrt{s})$ proves Eq. (B5).

There are $O(1/r)$ levels in $\mathcal{K}$, so its collar $|\lambda_j - t_0| \leq \eta$ has total $d_j = O(r/s)$. Moreover $t_j \leq 1 + Cr\mathrm{e}^{s\lambda_j} s^{-3/2}$, whose positive logarithm is at most $a\eta\mathcal{L} + o(\mathcal{L}) + C$. For $t_j \geq 1$ use $q_j(t_j - 1) \leq d_j$; for the other indices retain the $O(r^2)$ bound. Consequently

$$\limsup \frac{1}{r} \sum_{|\lambda_j - t_0| \leq \eta} q_j G(t_j) \leq C_{F,a}\eta. \tag{B8}$$

The constant is independent of small fixed η. This logarithmic estimate cannot be replaced by weak convergence alone.

3. Compact bands and activated final energy

Fix $t_0 < c < 4$, $M > 8$, and $\mathsf{P} = \mathbf{1}_{[c,M]}(B_\ell)$. If $a > 1$, choose $t_0 < e_0 < \min(c, 1)$; Eq. (A3) makes the full initial mass above e_0 be $o(r/s)$. Below it, Eq. (B7), a reduced resolvent analytic at zero, and Eq. (A6) give

$$\left\| \mathsf{P}\rho\mathsf{P} - \frac{r}{s}|v_\ell\rangle\langle v_\ell| \right\|_1 = o(r/s), \\ v_\ell = \frac{3}{m_1} \mathsf{P} B_\ell^{-1} (F f_{1,0,\ell}). \tag{B9}$$

The rest of the subthreshold inputs is exponentially paid by Eq. (A7). Strong resolvent convergence and absence of source atoms imply

$$\|v_\ell\|^2 \to \nu_F([c,M]), \qquad \langle v_\ell, B_\ell v_\ell \rangle \to \int_{[c,M]} E \nu_F(\mathrm{d}E). \tag{B10}$$

There are $O_M(1/r)$ band levels. Each final normalized eigenfunction has left sine coefficient $O(\sqrt{r})$ and, after compact transfer, a right piece bounded by $C\sqrt{r}(1 + x - d)$. This includes energy four: below it the right piece decays, at it it is linear, and above it $|\sin(kx)|/k \leq x$. Exponential decay of the zero-energy source then gives $|\langle \phi_j, v_\ell \rangle|^2 \leq C_{F,c,M} r$. The normalized source energy distribution has entropy $\mathcal{L} + O(1)$, between its min-entropy and log-rank bounds. Measurement and finite-alphabet entropy continuity transfer this to the actual conditional band distribution with error $o(\mathcal{L})$.

If $m = \sum_{[c,M]} p_j$, then $m \sim (r/s)\nu_F([c,M])$ and $-\log m = \mathcal{L} + o(\mathcal{L})$. Thus

$$-\sum_{[c,M]} p_j \log p_j = m[2\mathcal{L} + o(\mathcal{L})]. \tag{B11}$$

One logarithm is the conditional energy entropy and the other the rare total mass. With $\log Z_4 = \mathcal{L} + o(\mathcal{L})$ this gives

$$\frac{1}{r} \sum_{[c,M]} q_j G(t_j) \to \int_{[c,M]} (E - a^{-1}) \nu_F(\mathrm{d}E). \tag{B12}$$

The band Gibbs mass is $o(r/s)$, and replacing its KL sum by the nonnegative G sum changes it by only $O(r/s) = o(r)$.

For $a = 1$ choose $1 < c < 5/4$. The same argument includes all initial $E < 1$; keep $G_\ell =$

$Z_1^{-1}\mathsf{P}\mathbf{1}_{[1,\infty)}(A_\ell)\mathrm{e}^{-sA_\ell}\mathsf{P}$ separately. Activated input energy concentrates at one by Eqs. (A2)–(A3), and its mean squared mass away from the right plateau vanishes by Eq. (A8). The exact identity $(B_\ell-4)u_n = (E_n-1)u_n - 3(1-F)u_n$ therefore concentrates final energy at four. The fixed band retains all leading activated mass, and

$$\begin{aligned} \operatorname{Tr} G_\ell \sim \mathrm{e}^{-s}, \qquad & 0 \le G_\ell \le (\mathrm{e}^{-s}/Z_1)\mathsf{P}, \\ (s/r)\operatorname{Tr} G_\ell \to x, \qquad & (s/r)\operatorname{Tr} B_\ell G_\ell \to 4x. \end{aligned} \tag{B13}$$

The positive comparator $(r/s)|v_\ell\rangle\langle v_\ell| + G_\ell$ approximates the band covariance in trace norm by $o(r/s)$. Its normalized measured source weights are $O(r)$ and its background weights are $O(r\sqrt{s})$. Its conditional entropy is consequently still $\mathcal{L}+o(\mathcal{L})$; no source/background orthogonality is required. The mass coefficient is $\nu_F([c,M])+x$ and the energy coefficient is $\int_{[c,M]} E\nu_F(\mathrm{d}E) + 4x$. Repeating Eq. (B11) gives

$$\frac{1}{r}\sum_{[c,M]} q_j G(t_j) \to \int_{[c,M]} (E-1)\nu_F(\mathrm{d}E) + 3x. \tag{B14}$$

4. The entire final-energy tail

For initial $E<1$ and $M\ge 8$, spectral calculus and the exact overlap identity give

$$\begin{aligned} &\langle u_E, B\mathbf{1}_{[M,\infty)}(B)u_E\rangle \\ &= 9\langle Fu_E, \mathbf{1}_{[M,\infty)}(B)B(B-E)^{-2}Fu_E\rangle \\ &\le CM^{-1}\left\|Fu_E\right\|^2. \end{aligned} \tag{B15}$$

For $1\le E\le 2$ use Eq. (A18) instead, with $1-F$ replacing F. Inputs above two have prepared mean final energy $O(\mathrm{e}^{-2s})$, by heat-weight splitting, Eq. (A3), and $B = A+3F$. Thus

$$\begin{aligned} &\sum_{\lambda_j\ge M}\lambda_j p_j \\ &\le \frac{C_F r}{sM} + \frac{C_F \mathrm{e}^{-s}}{M}(s^{-1/2} + r\sqrt{s} + \mathrm{e}^{-s}) \\ &\quad + C_F\mathrm{e}^{-2s}. \end{aligned} \tag{B16}$$

For $t_j\ge 1$, $q_jG(t_j)\le p_j(s\lambda_j+\log Z_4)$; the other indices cost $O(r^2)$. Tail probability is at most tail energy divided by M. Bounded Ξ now gives

$$\limsup \frac{1}{r}\sum_{\lambda_j\ge M} q_j G(t_j) \le C_{F,a}/M. \tag{B17}$$

Partition the final spectrum below $t_0-\eta$, in its collar, between $t_0+\eta$ and M, and above M. All summands are nonnegative. Take the device limit first, then $\eta\downarrow 0$, then $M\to\infty$. Equations (B4), (B8), (B12) or (B14), and (B17) prove $P/r \to \int (E-a^{-1})_+\nu_F(\mathrm{d}E)+3x$. The finite first moment permits exhaustion. Subtracting this result from Eq. (B2) proves Eq. (24).

Appendix C: The sharp-wall infrared work contribution

We prove Eqs. (30)–(32) under the basic regime with $\Xi = O(1)$. In this appendix $F = \mathbf{1}_{x>0}$, $N = \ell/\sqrt{s}$, and $O_{nj} = \langle \phi_j, u_n\rangle$.

1. Exact low-level overlaps

The actual subbarrier Sturm labels obey

$$\begin{aligned} \ell k_n + \delta_1(k_n) = n\pi, \qquad & \ell q_j + \delta_4(q_j) = j\pi, \\ \delta_h(k) = \arctan\frac{k\tanh(\ell\sqrt{h-k^2})}{\sqrt{h-k^2}}. & \end{aligned} \tag{C1}$$

On fixed small compact energy intervals the normalized left pieces are $a_h(k)\sin(k(x+\ell))$, with $a_h(k) = \sqrt{2r}[1+O(r)]$ and $\sin\delta_h(k) = (k/\sqrt{h})[1+O(\mathrm{e}^{-b_0\ell})]$. The right pieces are the interface values times $\sinh(\sqrt{h-k^2}(\ell-x))/\sinh(\ell\sqrt{h-k^2})$. Their cross integral equals $[\sqrt{1-k^2}+\sqrt{4-q^2}]^{-1} + O(\ell\mathrm{e}^{-b_0\ell})$. Hence, for $j\ne n$,

$$\begin{aligned} |q_j^2-k_n^2|\,|O_{nj}|^2 = \frac{r^2k_n^2q_j^2}{|q_j^2-k_n^2|} \\ [1+O(k_n^2+q_j^2+r+\ell\mathrm{e}^{-b_0\ell})]. \end{aligned} \tag{C2}$$

The numerator before dividing by the gap is $3\langle\phi_j, Fu_n\rangle = \pm rk_nq_j[1+O(k_n^2+q_j^2+r+\ell\mathrm{e}^{-b_0\ell})]$: the leading factor is $3(2r)k_n(q_j/2)/(1+2)$.

Put $m=j-n$. Subtracting the exact phases gives

$$\begin{aligned} (\ell+\delta_4'(\xi))(q_j-k_n) &= \pi m + d(k_n), \\ d(k) &= \delta_1(k)-\delta_4(k) \\ &= k/2 + O(k^3 + k\mathrm{e}^{-b_0\ell}). \end{aligned} \tag{C3}$$

For $m\ne 0$ and $k/2\le q_j\le 2k$, this implies $|q_j-k|^{-1} = \ell(\pi|m|)^{-1}[1+O(k/|m|+r)]$. Since $q^2/(q+k) = k/2 + O(|q-k|)$, each near-index term in Eq. (C2) has leading value $rk^3/(2\pi|m|)$. The two index ranges have sizes $\ell k/\pi + O(1)$ and $\ell k/(2\pi)+O(1)$, so their harmonic sums total $2\log(\ell k)+O(1)$. An empty lower range for the first few modes is harmless: then ℓk is bounded and is at least $\pi/2$.

For all sufficiently small $0<\delta<\delta_0$ and $k^2<\delta/16$, the uniform result is

$$\begin{aligned} &\sum_{j\ne n,\ q_j^2<\delta} |q_j^2-k^2||O_{nj}|^2 \\ &= \frac{rk^3}{\pi}\log(\ell k) + R_{n,\delta}, \\ |R_{n,\delta}| &\le Cr\{k^2\sqrt{\delta} + k^3 + k^5\log(2+\ell k) \\ &\quad + (r+\ell\mathrm{e}^{-b_0\ell})k^3\log(2+\ell k)\}. \end{aligned} \tag{C4}$$

The phase correction costs $O(rk^4\sum m^{-2})$. Replacing $q^2/(q+k)$ costs $O(r^2k^2(\ell k+1)) = O(rk^3)$, because

$k \geq \pi r/2$. The relative k^2 correction multiplies the harmonic sum and produces the k^5 term. In the far ranges, $q < k/2$ costs $O(rk^3)$ since $q^2/|q^2-k^2| \leq Cq^2/k^2$; $2k < q < \sqrt{\delta}$ costs $O(rk^2\sqrt{\delta})$ by counting levels. This proves Eq. (C4) without integrating across a discrete spectral pole.

2. Every thermal sector and additive error control

For every fixed $d_0 < 1$, the complete conditional uncapped mean obeys

$$\sum_j |\lambda_j - E_n||O_{nj}|^2 \leq C_{d_0} r[k_n^2 + k_n^3 \log(2+\ell k_n)] \quad (E_n \leq d_0). \tag{C5}$$

Below a fixed final cutoff less than four, use the numerator bound $Crkq$, the spacing $|q_j - k_n| \geq \pi r(|j-n|-1/2)$, the same near/far split, and the paired shift $O(rE_n)$. Above that cutoff the gap is separated from zero; $|\omega| \leq C\omega^2$ and

$$\sum_j (\lambda_j - E_n)^2 |O_{nj}|^2 = 9\,\|Fu_n\|^2 \tag{C6}$$

pay its entire infinite range by CrE_n.

Equation (A7) bounds $\sum_{1/2 \leq E_n < 1} \|Fu_n\|^2$ uniformly, and $N_A(1) = O(\ell)$. By Cauchy–Schwarz, the conditional absolute first moment is at most $3\,\|Fu_n\|$. Thus the complete collar $[d_0, 1)$, $d_0 > 1/2$, costs after Gibbs normalization and scaling at most

$$Cs^{3/2}\sqrt{\ell}\mathrm{e}^{-sd_0} \leq Cs\mathrm{e}^{-s(d_0-1/2)} = o(1), \tag{C7}$$

because bounded $\Xi = s\ell\mathrm{e}^{-s}$ implies $\ell \leq C\mathrm{e}^s/s$. For $s^{-1/2} \leq E_n < d_0$, Eq. (C5) and a Gaussian heat-weight split give a polynomial in s times $\mathrm{e}^{-\sqrt{s}/2}$. The count's factor ℓ cancels against Z, and $\log \ell = O(s)$; this intermediate sector is also $o(1)$.

For activated inputs, Eqs. (A2) and (A10) imply $T/Z \sim \mathrm{e}^{-s}$ and $T_-/T = O(r\sqrt{s} + s^{-1/2})$. Both $h_c(w) = \min(|w|, c)$ and $h_\infty(w) = |w|$ are one-Lipschitz. Equation (A18) and Cauchy–Schwarz therefore give

$$\Big| \sum_{E_n \geq 1} \mathrm{e}^{-sE_n} \sum_j h_c(\lambda_j - E_n)|O_{nj}|^2 - \min(3,c)T \Big| \leq 3\sqrt{TT_-}. \tag{C8}$$

Its scaled error is $O(\Xi\sqrt{r\sqrt{s} + s^{-1/2}}) = o(1)$, and its main contribution is $\min(3,c)\Xi$; for uncapped work replace the coefficient by three. Every activated input and final energy is included in this estimate.

Only $E_n < s^{-1/2}$ now remains. Gaussian Stieltjes summation gives

$$\begin{aligned} s\langle E_n \mathbf{1}_{E_n < s^{-1/2}} \rangle_\rho &\to \tfrac{1}{2}, \\ s\langle E_n^2 \mathbf{1}_{E_n < s^{-1/2}} \rangle_\rho &\to 0, \\ \frac{s}{\pi}\langle k_n^3 \log(\ell k_n) \mathbf{1}_{E_n < s^{-1/2}} \rangle_\rho &= \frac{\log N}{\pi^{3/2}\sqrt{s}} + o(1). \end{aligned} \tag{C9}$$

For the logarithmic moment set $v = \sqrt{s}\,k$. Its continuum numerator is $(\ell/\pi)s^{-2}\int_0^\infty v^3 \mathrm{e}^{-v^2}(\log N + \log v)\mathrm{d}v$. The first integral is $1/2$, the second finite. The test's total variation is $O(s^{-3/2}(1+\log N))$, giving an additive scaled quadrature error $O(r(1+\log N)) = o(1)$. The relative partition error $O(r\sqrt{s} + \mathrm{e}^{-s})$ costs only $O(r\log N + \mathrm{e}^{-s}\log N/\sqrt{s}) = o(1)$ against the possibly diverging main term. The moving Gaussian tail vanishes because $\log \ell = O(s)$. Unnormalized fixed-cutoff heat errors carry an ℓ prefactor; their normalized contributions are the errors asserted to vanish here.

The thermal averages of the remainders in Eq. (C4) obey

$$\begin{aligned} s\langle k^2\sqrt{\delta}\rangle &= O(\sqrt{\delta}), \\ s\langle k^3\rangle &= O(s^{-1/2}), \\ s\langle k^5 \log(2+\ell k)\rangle &= O(\log \ell / s^{3/2}) + o(1) = o(1). \end{aligned} \tag{C10}$$

The last remainder also vanishes, using $r\sqrt{s} \to 0$, $\sqrt{s} = o(\ell)$ and exponential decay in ℓ. These estimates are additive even when $\log N/\sqrt{s}$ diverges.

3. Paired shift, source matching and closure

At $m = 0$, Eq. (C3) gives

$$\begin{aligned} \lambda_n - E_n &= rE_n[1 + O(E_n + r + \mathrm{e}^{-b_0\ell})], \\ |O_{nn}|^2 &= 1 + O(E_n + r). \end{aligned} \tag{C11}$$

For the second bound choose signs so the left sine pieces agree: their normalized difference has norm $O(r + \ell|q_n - k_n|) = O(r + k_n)$; their right norms are $O(\sqrt{r}\,k_n)$. The paired contribution consequently tends to $1/2$ by Eq. (C9).

Fix $0 < \delta < \min(\delta_0, c)$. The remaining final modes below δ contribute $\log N/(\pi^{3/2}\sqrt{s}) + R_\delta$, with $\limsup |R_\delta| \leq C\sqrt{\delta}$. For all final energies above δ use the bounded reduced spectral function

$$G_{E,\delta}(z) = \mathbf{1}_{z \geq \delta} \frac{h_c(z-E)}{(z-E)^2}. \tag{C12}$$

For $E < \delta/2$ it is uniformly bounded and differs from $G_{0,\delta}$ by $O_\delta(E)$ in supremum norm. For uncapped work it is simply $\mathbf{1}_{z\geq\delta}/(z-E)$. Exact matching gives $Fu_n/(\sqrt{2r}k_n) \to f$ in L^2, uniformly over the moving low-input window. The overlap identity implies

$$\begin{aligned} \sum_{\lambda_j \geq \delta} h_c(\lambda_j - E_n)|O_{nj}|^2 &= 9\langle Fu_n, G_{E_n,\delta}(B_\ell)Fu_n\rangle \\ &= 2rE_n[\Phi_\delta(c) + o(1)], \\ \Phi_\delta(c) &= \int_{[\delta,\infty)} \min(E,c)\nu(\mathrm{d}E). \end{aligned} \tag{C13}$$

Strong resolvent convergence applies at the cutoff because the source has no atom there; the bounded function

includes the whole infinite final range. Thermal averaging gives $\Phi_\delta(c)$. The omitted source first moment is $O(\sqrt{\delta})$. Combine these three disjoint final sectors and the preceding input bounds, take the device limit, then send $\delta \downarrow 0$. This proves Eq. (30). The same proof, directly with the uncapped test and $\int E\nu(\mathrm{d}E) = 1$, proves Eq. (31).

The exact signed mean is $3\,\mathrm{Tr}\,\rho F$. Its low and activated contributions are $3r/(2s)$ and $3\mathrm{e}^{-s}$, respectively, proving the signed formula in Eq. (32). Subtract it from the absolute-work result. As an independent sign check, the negative near-index sum uses just $m < 0$, half the harmonic coefficient; far negative terms cost $O(rk^3)$, and the activated negative mean is $o(1)$ after scaling since $(-w)_+$ is one-Lipschitz and vanishes at three. Finally, when $s/\mathcal{L} \to a > 1$, $\log N/\sqrt{s} \sim \sqrt{s}/a$ and $\Xi \to 0$, which proves both ratios in Eq. (33).

Appendix D: Source reconstruction and frequency rejection

1. Positive transform, finite window and frequency rejection

For a positive measure on $(0,\infty)$ with first moment $M < \infty$, $C(t)$ and $\Phi(c)$ are finite and $|C'(t)| \le M/2$ by dominated differentiation. Integration by parts and the Dirichlet sine integral give $\int_0^\infty (1-\cos ut)t^{-2}\mathrm{d}t = \pi|u|/2$. Product-to-sum consequently gives

$$\int_0^\infty \frac{(1-\cos ct)(1-\cos Et)}{t^2}\mathrm{d}t = \frac{\pi}{2}\min(c,E). \tag{D1}$$

Tonelli applied to the nonnegative integrand proves Eq. (34). Its kernel has mass $2c$, so the uniform-additive-error constant $2c$ is exact.

If $\nu'(E) \le KE^{-3/2}$ for $0 < E < E_*$, then $C(t) \le K\sqrt{\pi/2}\sqrt{t} + M/E_*$. Therefore the positive omitted time tail satisfies

$$0 \le \Phi(c) - \Phi_T(c) \le \frac{16K}{\sqrt{2\pi T}} + \frac{8M}{\pi E_* T}. \tag{D2}$$

For exact midpoint weights on a partition of maximum mesh Δ, source sample errors at most ε give

$$\left|\sum_i w_i \widehat{C}(t_i) - \Phi(c)\right| \le \frac{16K}{\sqrt{2\pi T}} + \frac{8M}{\pi E_* T} + 2c\varepsilon + \frac{cM\Delta}{2}. \tag{D3}$$

The last term uses $|C'| \le M/2$, a maximal midpoint distance $\Delta/2$, and $\sum w_i \le 2c$.

These constants can be explicit for all fixed-width profiles. Let $n_d = 2d + 1/2$, $m_* = (2d+1)^{-1}$ and

$$E_* = \min\{1/4, \gamma_d/4, m_*/(8n_d)\}, \qquad K = \frac{144 n_d^2}{\pi m_*^4}. \tag{D4}$$

Equation (A4) implies $\|f_{4,E}\| \le 2\sqrt{n_d}$ and $m_4(E) \ge m_*/2$ on this interval; also $\|g\| \le \sqrt{n_d}$. Substitution into Eq. (A13) proves the stated threshold bound. The zero-energy energy identity gives $\dot{m}_1 \le m_1$, hence $M_F \le 3/m_*$.

For a sharper subsequent tail, if $|\nu'(E) - kE^{-3/2}| \le BE^{-1/2}$ below E_*, the signed difference from $kE^{-3/2}\mathrm{d}E$ on the entire positive axis has total variation at most $R_0 = 2B\sqrt{E_*} + M/E_* + 2k/\sqrt{E_*}$. Thus $C(t) = A\sqrt{t} + R(t)$, $A = k\sqrt{\pi/2}$, $|R(t)| \le R_0$. An integration by parts of the cosine tail yields

$$\left|\Phi(c) - \Phi_T(c) - \frac{8A}{\pi\sqrt{T}}\right| \le \frac{8A}{\pi c T^{3/2}} + \frac{8R_0}{\pi T}. \tag{D5}$$

For the actual source $A = D_F^2/\sqrt{2\pi}$; the remainder is not uniform as the cap tends to zero.

Exact all-time source data determine all caps, then the measure via $\Phi'_+(c) = \nu((c,\infty))$ and $\Phi'_-(c) = \nu([c,\infty))$, by dominated one-sided difference quotients away from zero. This is not a stable inversion in a strong measure norm. In particular adding $\varepsilon\delta_H$ changes C uniformly by at most ε but changes its first moment by εH. Resonance aliasing in the main text therefore does not contradict all-time injectivity. It can even preserve two moments with smooth densities: take a smooth probability density ψ supported in $(0,c)$, of mean $c/2$, and a common base $kE^{-3/2}\mathrm{e}^{-E}\mathrm{d}E$. Adding respectively $\beta(3T_3\psi + T_9\psi)/4$ and $\beta(\psi + 3T_6\psi)/4$, where T_j translates by j, gives equal first and second moments and identical resonant samples, but caps differing by $\beta c/8$. These are measure-class examples, not an asserted pair of realizable wall profiles.

For the actual finite work probability, Eq. (D1) also proves the exact all-time identity stated after Eq. (34). For fixed $0 < \eta < c$ put $h_{\eta,c}(w) = \min(|w|, c) - \min(|w|, \eta)$. Its quotient by w^2, set to zero at zero, is bounded continuous with norm at most $1/(4\eta)$. Equation (29) therefore proves Eq. (35). Splitting the source cap at η and E_* gives the cutoff budget

$$0 \le \Phi_F(\eta) \le 4K\sqrt{\eta} + \eta M_F/E_* \quad (\eta \le E_*). \tag{D6}$$

Subtracting two absolutely integrable cosine identities gives its signed time kernel. Its total absolute weight is at most $2(c+\eta)$; since $p_{r,s} \le 1$, the scaled actual-device tail beyond T is at most $8(s/r)/(\pi T)$. The ordered limiting filter does not supply an uncharged finite observation protocol.

Appendix E: Resonant source selection and complete entropy bounds

We prove the actual conditional-state assertion, including its infinite-dimensional entropy limit. All operators in this appendix are in the boundary coordinate on $(-\ell, \ell)$, with $\ell = 1/r$. The resonance $\tau = 2\pi n/3$ is fixed, where n is a positive integer. Constants may depend on this resonance and on a fixed smooth profile, but not on ℓ or s. For a density operator σ and $q > 0$, $q \ne 1$, the Rényi entropy is $S_q(\sigma) = \log \mathrm{Tr}\,\sigma^q/(1-q)$; the order-one entropy is $S(\sigma) = -\mathrm{Tr}\,\sigma\log\sigma$.

1. The sharp low-energy covariance

For the sharp wall, put $Q = \mathbf{1}_{(0,\ell)}$ and $A = -\partial_x^2 + Q$, $B = -\partial_x^2 + 4Q$. At $E = k^2 < 1$ define

$$\kappa = \sqrt{1-E}, \qquad d(E) = \kappa \coth(\kappa\ell),$$
$$f_{E,\ell}(x) = \frac{\sinh[\kappa(\ell - x)]}{\sinh(\kappa\ell)} \mathbf{1}_{(0,\ell)}(x). \tag{E1}$$

Matching the left sine and right hyperbolic sine gives $k\ell + \arctan[k/d(k^2)] = j\pi$. If u_E is normalized and $b_E = u_E(0)$, direct integration gives

$$Qu_E = b_E f_{E,\ell}, \qquad |b_E|^2 = \frac{2E}{D_\ell(E)},$$
$$D_\ell(E) = \ell + [\ell - 2\ell E]\,\mathrm{csch}^2(\kappa\ell) + \frac{\coth(\kappa\ell)}{\kappa}. \tag{E2}$$

On $0 \le E \le 1/2$, one has $\ell \le D_\ell(E) \le \ell + 3$ for large ℓ. The phase is increasing and its count differs from $\ell k/\pi$ by at most one. Consequently, for each fixed integer $j \ge 0$,

$$M_j := \sum_{E<1/2} |b_E|^2 \mathrm{e}^{-sE} E^j = \frac{\Gamma(j+3/2)}{\pi} s^{-j-3/2}[1+o(1)]. \tag{E3}$$

Indeed, Stieltjes integration of $k^{2j+2}\mathrm{e}^{-sk^2}$ against the count gives discrepancy at most twice its maximum. After multiplication by $2/\ell$, its relative size is $O_j(\sqrt{s}/\ell)$. The normalization replacement costs $O(\ell^{-1})$, and the omitted Gaussian integral above $k = 1/\sqrt{2}$ is exponentially small. These errors vanish in Eq. (5). Write $m_s = (2\sqrt{\pi}s^{3/2})^{-1}$, so $M_0 \sim m_s$.

Duhamel's formula gives the exact column identity

$$K_t u_E = b_E \mathrm{e}^{-itE} V_\ell(E), \qquad V_\ell(E) = \frac{3}{2} h_t(B-E) f_{E,\ell}. \tag{E4}$$

The low accepted covariance is thus

$$S = \sum_{E<1/2} |b_E|^2 \mathrm{e}^{-sE} |V_\ell(E)\rangle\langle V_\ell(E)|. \tag{E5}$$

Different initial eigenstates contribute positive outer products; their amplitudes are not coherently averaged.

Let T_ℓ be the Dirichlet inverse of $-\partial_x^2 + 1$ on $(0,\ell)$. The inhomogeneous right boundary problem gives $f_{E,\ell} = (I - ET_\ell)^{-1} f_{0,\ell}$, with $\|T_\ell\| \le 1$ and $\|f_{0,\ell}\| \le 1/\sqrt{2}$. Using $h_t(B-E) = -i\int_0^t \mathrm{e}^{-iuB}\mathrm{e}^{iuE}\mathrm{d}u$, the Taylor coefficients of $V_\ell(E) = \sum_{j\ge 0} E^j V_{j,\ell}$ obey

$$\|V_{j,\ell}\| \le M_t := \frac{3}{2\sqrt{2}} \int_0^t \mathrm{e}^u \mathrm{d}u. \tag{E6}$$

Therefore $\|S - M_0 |V_\ell(0)\rangle\langle V_\ell(0)|\|_1 \le C_t M_1$. Increasing-interval Dirichlet form convergence implies strong resolvent convergence to $B_\infty = -\partial_x^2 + 4\mathbf{1}_{x>0}$ and hence strong propagator convergence at every fixed time. Since $f_{0,\ell} \to f = \mathrm{e}^{-x}\mathbf{1}_{x>0}$ in L^2,

$$V_\ell(0) \longrightarrow v_t = \frac{3}{2} h_t(B_\infty) f, \qquad \|v_t\|^2 = C(t) > 0. \tag{E7}$$

For positivity, B_∞ has no eigenvalues: positivity excludes negative ones, and at nonnegative energy its free left solution is affine or oscillatory and cannot be square integrable unless zero. ODE uniqueness then makes the whole solution zero. Its spectral measure for nonzero f is atom-free, whereas the real zero set of h_t is countable. This proves the strict inequality in Eq. (E7). The actual comparison projection is $\Pi_{r,t} = |V_\ell(0)\rangle\langle V_\ell(0)| / \|V_\ell(0)\|^2$.

2. An all-energy boundary estimate at resonance

Let $H = -\partial_x^2 + V$, $0 \le V \le v$, on an interval with a unit subinterval adjacent to the point under consideration. For $R = (H+1)^{-1}$, spectral calculus and the differential equation give

$$\begin{aligned} \|Rg\| &\le \|g\|, \\ \|(Rg)'\| &\le \tfrac{1}{2}\|g\|, \\ \|(Rg)''\| &\le (v+2)\|g\|. \end{aligned} \tag{E8}$$

The elementary endpoint inequality $|h(0)|^2 \le 2\|h\|^2 + \|h'\|^2$ on a unit interval, applied to Rg and $(Rg)'$, gives by Parseval

$$\sum_E \frac{|u_E(0)|^2 + |u_E'(0)|^2}{(1+E)^2} \le D_v := \frac{11}{4} + (v+2)^2. \tag{E9}$$

This also defines the bounded resolvent-smoothed distributions $R\delta_0, R\delta_0'$. The constant is independent of total length, local level spacing, and threshold proximity. Separated remote Neumann conditions give the same estimate.

At resonance the function

$$q_\tau(z) = \frac{\mathrm{e}^{-i\tau z} - 1}{z(z-3)} \tag{E10}$$

is entire. Green's identity on the controlled interval gives, for every actual initial and final eigenstate,

$$\begin{aligned} (\lambda - E)\langle \phi_\lambda, u_E\rangle &= 3\langle \phi_\lambda, Qu_E\rangle, \\ (\lambda - E - 3)\langle \phi_\lambda, Qu_E\rangle &= -\overline{\phi_\lambda(0)} u_E'(0) + \overline{\phi_\lambda'(0)} u_E(0). \end{aligned} \tag{E11}$$

Physical outer endpoint terms vanish by the Dirichlet conditions. Combining the identities, including their removable equal-energy values, proves

$$K_\tau u_E = -\frac{3}{2}\mathrm{e}^{-i\tau E} q_\tau(B-E)[u_E'(0)\delta_0 + u_E(0)\delta_0']. \tag{E12}$$

Its distributional meaning is fixed by $q_\tau(B-E)\delta_0^{(i)} = [q_\tau(B-E)(B+1)](B+1)^{-1}\delta_0^{(i)}$, whose bracket is bounded.

Near zero use $|e^{-i\tau x}-1|\le\tau|x|$; near three use the resonant version of that estimate; away from both points use the numerator bound two. These give $\sup_{x\in\mathbb{R}}(1+|x|)|q_\tau(x)|\le 5\tau/2$. Since $1+\lambda\le(1+E)(1+|\lambda-E|)$, the input and output versions of Eq. (E9) imply, for each $d_0>0$,

$$\begin{aligned}&\operatorname{Tr} K_\tau \mathbf{1}_{[d_0,\infty)}(A)e^{-sA}K_\tau^*\\&\quad\le C_\tau \sup_{E\ge d_0} e^{-sE}(1+E)^4 \le C_\tau(1+d_0)^4e^{-sd_0},\end{aligned}\tag{E13}$$

where the last inequality holds for $s\ge 4/(1+d_0)$. Taking $d_0=1/2$ pays the complete complement of Eq. (E5), including every threshold and high mode. No box-volume factor remains.

3. All-index tails and entropy

The first m Taylor vectors in Eq. (E6) leave error at most $2M_\tau E^m$ on $0\le E\le 1/2$. Positivity and the variational characterization of eigenvalue sums yield

$$\sum_{j>m}\lambda_j(S)\le 4M_\tau^2M_{2m}.\tag{E14}$$

The actual cutoff gives $E^{2m}\le 4^{-(m-1)}E^2$, so Eqs. (E3) and (E7) imply

$$\sum_{j>m}\lambda_j(S/\operatorname{Tr}S)\le A_\tau s^{-2}4^{-(m-1)},\qquad m\ge 1.\tag{E15}$$

Summing $-\lambda\log\lambda$ under this envelope, including the $O(s^{-2})$ deficit of the leading eigenvalue, proves $S(S/\operatorname{Tr}S)=O_\tau(s^{-2}\log s)$. For each fixed $0<q<1$ it also proves $\operatorname{Tr}(S/\operatorname{Tr}S)^q=1+O_{\tau,q}(s^{-2q})$.

Write $G=K_\tau\mathbf{1}_{[1/2,\infty)}(A)e^{-sA}K_\tau^*$. The two vector curves $g_i(E)=q_\tau(B-E)\delta_0^{(i)}$, $i=0,1$, are norm-analytic on the strip $|\operatorname{Im}E|\le 1$ and satisfy

$$\|g_i(E)\|\le A_\tau(1+|E|).\tag{E16}$$

Indeed the entire divided difference bounds q_τ near its removable points, while its quotient gives quadratic decay at large real part; hence $(1+|z|)|q_\tau(z)|$ is bounded on the strip. Factoring through the local resolvent vectors proves Eq. (E16).

In the bin $[1/2+j,3/2+j)$ expand both curves at its midpoint on the radius-one circle. Retaining m terms costs at most $2m$ dimensions and has vector error $A_\tau(j+3)2^{-m}$. Equation (E9) bounds the sum of squared input boundary coefficients in the bin by $D_1(j+3)^2$. Retaining their Cauchy–Schwarz cross bound, the trace outside the approximation space is at most

$$A_\tau(j+3)^4e^{-s(1/2+j)}4^{-m}.\tag{E17}$$

Choose $m_j=M-j$ for $0\le j<M$ and none above. The total dimension is at most $M(M+1)$. Both the retained-bin errors and the omitted bins sum because $4e^{-s}<1$ for $s\ge 2$. Thus

$$\begin{aligned}\sum_{k>M(M+1)}\lambda_k(G)&\le A_\tau e^{-s/2}4^{-M},\\ \lambda_k(G)&\le A'_\tau e^{-s/2}e^{-c\sqrt{k}}.\end{aligned}\tag{E18}$$

This controls both complete input and final spectral tails.

Let $T=\operatorname{Tr}(S+G)\sim C(\tau)m_s$ and $Y=G/T$. Its eigenvalues are bounded by $\varepsilon_s e^{-c\sqrt{k}}$ with $\varepsilon_s=O_\tau(s^{3/2}e^{-s/2})$. The monotonicity of $-x\log x$ below $1/e$ and summability of this envelope show

$$\begin{aligned}S(Y)&=O_\tau(s^{5/2}e^{-s/2}),\\ \operatorname{Tr}Y^q&=O_{\tau,q}(\varepsilon_s^q),\quad 0<q<1.\end{aligned}\tag{E19}$$

Here $S(X)=-\operatorname{Tr}X\log X$ is also used for subnormalized positive operators. Positive-sum entropy bounds give

$$S(\sigma_\tau)=O_\tau(s^{-2}\log s+s^{5/2}e^{-s/2})\to 0.\tag{E20}$$

For $0<q<1$ use trace-power subadditivity; for $q>1$ the largest eigenvalue tending to one suffices. This proves every fixed positive Rényi limit. Finally, $Z=(1+e^{-s})\ell/(2\sqrt{\pi}\sqrt{s})+O(1)$ by Dirichlet/Neumann splitting. Division of the accepted trace by this complete partition proves $p_\tau\sim C(\tau)r/s$.

4. Fixed smooth plateau profiles

We supply the complete smooth high-sector argument; smooth entropy is not inferred from the sharp trace bound. Suppose $F=0$ to the left of $-d$ and $F=1$ to the right of d, as in the main text. The right Dirichlet gap, boundary normalization, and moments are proved in Eqs. (A4)–(A6). For explicit analytic radii one can take

$$\delta_F=\min\left\{\frac14,\frac{\gamma_d}{4},\frac{1}{8(2d+1)(2d+1/2)}\right\}>0.\tag{E21}$$

The zero-energy right solution satisfies $\|f_{1,0,\ell}\|^2\le 2d+1/2$, and the resolvent series gives $\|f_{1,E,\ell}\|\le 2\sqrt{2d+1/2}$ for $|E|\le 2\delta_F$. The exact source curve is

$$V_\ell(E)=\frac32 h_\tau(B-E)Ff_{1,E,\ell}.\tag{E22}$$

It is uniformly norm-analytic on that disk. Cauchy's estimate on radius $2\delta_F$, evaluated for $E\le\delta_F$, gives the same normalized geometric low tail as Eq. (E15). The boundary moment has the factor m_1^{-2}, so its accepted trace is asymptotic to $C_F(\tau)m_s$.

For the complete high sector split $F=F_c+Q_d$, where $F_c=F\mathbf{1}_{(-d,d)}$ and $Q_d=\mathbf{1}_{(d,\ell)}$. An exact all-energy

identity is

$$K_\tau u_E = \frac{3}{2}\mathrm{e}^{-i\tau E}\big\{h_\tau(B-E)F_c u_E - q_\tau(B-E)[u'_E(d)\delta_d + u_E(d)\delta'_d]\big\}. \tag{E23}$$

The interface distributions result from the artificial split, not a discontinuity of F. The plateau identity $(B-E-3)Q_d u_E = -u'_E(d)\delta_d - u_E(d)\delta'_d$, multiplied by the removable resonant quotient, proves the formula.

On $[-d,d]$ write $u_E = u_E(d)\phi_E + u'_E(d)\psi_E$, where the two solutions have terminal data $(1,0)$ and $(0,1)$ at d. Their ODE integral series is entire in E, and first-order Gronwall gives a bound $\exp[2d(2+|E|)]$ on each solution and derivative. Thus Eq. (E23) is $\mathrm{e}^{-i\tau E}$ times two norm-analytic vectors with the actual boundary coefficients. On the radius-one circle centered at $\delta_F + j + 1/2$, their norms are at most $C_{d,\tau}(j+3)\mathrm{e}^{2dj}$. The local trace inequality at d is unchanged. A bin with m Taylor terms consequently has trace error at most

$$C_{d,\tau}(j+3)^4 \mathrm{e}^{-s\delta_F}\mathrm{e}^{-(s-4d)j}4^{-m}. \tag{E24}$$

The same allocation $m_j = M - j$ proves, for $s \geq 4d+2$,

$$\sum_{k>M(M+1)} \lambda_k(G_F) \leq C_{d,\tau}\mathrm{e}^{-s\delta_F}4^{-M}, \tag{E25}$$

where G_F includes every input energy at least δ_F. In particular it has the all-index stretched-exponential envelope needed after conditioning. Repeating the normalization and entropy argument gives

$$S(\sigma_\tau) = O_{F,\tau}(s^{-2}\log s + s^{5/2}\mathrm{e}^{-s\delta_F}) \to 0, \tag{E26}$$

and all fixed positive Rényi limits. The positive source limit follows from strong resolvent convergence and the absence of eigenvalues of the full-line operator, whose left half-line is free. Its coefficient is precisely $\|(3/2)h_\tau(B_\infty)Ff_1\|^2/m_1^2 = C_F(\tau)$. These constants concern a fixed profile in penetration-length units; the theorem does not keep a positive physical transition width fixed as $r \to 0$.

Appendix F: One controlled wall versus two in the same Gibbs well

Use the connected initial well in Eq. (18). Restriction to the right half, multiplied by $\sqrt{2}$, is unitary from each initial parity subspace. In right-interface coordinates the half interval is $(-a,\ell)$, $a = 1/(2r)$, $\ell = 1/r$. Even parity gives a Neumann condition at $-a$; odd parity gives a Dirichlet condition. This decomposition is an exact representation of the initial Sturm problem and introduces no physical midpoint wall.

For $E = k^2 < 1$, the quantization conditions are

$$ka + \arctan[k/d(E)] = \begin{cases} (j-1/2)\pi, & \text{even}, \\ j\pi, & \text{odd}, \end{cases} \tag{F1}$$

with the right function in Eq. (E1). Integration gives the normalized half-state boundary amplitude

$$b_E^2 = \frac{2E}{a + [a-(a+\ell)E]\,\mathrm{csch}^2(\kappa\ell) + \coth(\kappa\ell)/\kappa}. \tag{F2}$$

The denominator lies between a and $a+3$ for $E \leq 1/2$ and large a. Each parity count differs from ak/π by at most one. The proof of Eq. (E3) therefore gives

$$M_{j,+} \sim M_{j,-} \sim \frac{\Gamma(j+3/2)}{\pi}s^{-j-3/2}. \tag{F3}$$

Equal leading weights follow from this computation, not reflection symmetry alone.

When only the right wall is controlled, the actual full-state boundary amplitude is $\beta_E = b_E/\sqrt{2}$. Its union moment is $(M_{j,+}+M_{j,-})/2$, with the single-wall leading coefficient. The final operator is on $(-2\ell,\ell)$ with height one on $(-2\ell,-\ell)$ and height four on $(0,\ell)$. It does not preserve initial parity. For both initial parity sectors,

$$K_R u_E = \beta_E \mathrm{e}^{-i\tau E}\frac{3}{2}h_\tau(B_R - E)f_{E,\ell}. \tag{F4}$$

Thus there is one common source curve. Both initial parity populations approach the same output vector. The uncontrolled left barrier and the endpoints recede in the source limit, giving the same full-line v_τ as Eq. (E7). Only the controlled interface contributes to Eq. (E12); the uncontrolled left wall commutes with Q_R. The preceding local trace, analytic-curve, and entropy bounds apply unchanged. Hence the accepted trace is $C(\tau)m_s[1+o(1)]$ and the output entropy tends to zero.

When both walls are controlled, both final parity blocks are exactly invariant and orthogonal. Their half-interval operators are the same sharp A, B pair; only the remote midpoint condition differs. At a negative spectral parameter the reflected Green terms from a remote Neumann or Dirichlet endpoint decay exponentially, giving the same strong source limit for both. Local resolvent estimates and their integrations by parts hold with either remote condition. Each block therefore has accepted trace $C(\tau)m_s[1+o(1)]$ and normalized entropy tending to zero. Their weights approach $1/2$. If the weight is z, the exact identity

$$S(z\sigma_+ \oplus (1-z)\sigma_-) = h_2(z) + zS(\sigma_+) + (1-z)S(\sigma_-) \tag{F5}$$

where $h_2(z) = -z\log z - (1-z)\log(1-z)$, proves the limit $\log 2$. The block trace-power identity proves the same limit for each fixed positive Rényi order.

For the unchanged input, three-interval bracketing gives

$$Z = \frac{1+2\mathrm{e}^{-s}}{2\sqrt{\pi}r\sqrt{s}} + O(1). \tag{F6}$$

It proves the two stated probabilities and their ratio two, as well as input purity $\sim \sqrt{2\pi}r\sqrt{s}$. There is no finite-parameter identity $K_2 = K_L + K_R$, and the calculation does not assert that arbitrary interface counts equal accepted rank.

Appendix G: Survival of repeated noncommuting heralds

Here the resonance is the fundamental one, $\tau = 2\pi/3$. We prove injectivity for the actual full-line difference of propagators and then transfer the complete thermal estimates. This is needed because the difference of two unitaries can be nonnormal and can even have a zero square; a scalar spectral filter argument does not apply to K^m.

1. The sharp step

Suppose $Kg = 0$, $\|g\| = 1$. Set $c = 3/2$, $W(u) = \mathrm{e}^{iuA}\mathrm{e}^{-iuB}$, $Z(u) = \mathrm{e}^{icu}W(u)$, and $S(u) = \mathrm{e}^{iuA}(2Q - I)\mathrm{e}^{-iuA}$. Bounded perturbation makes $Z(u)g$ strongly continuously differentiable, with

$$Z'(u) = -icS(u)Z(u), \qquad S(u)^* = S(u), \quad S(u)^2 = I. \tag{G1}$$

On the real unit sphere of the complex Hilbert space, the curve $\psi(u) = Z(u)g$ has speed c, joins g to $-g$, and has length $c\tau = \pi$. It is a shortest great semicircle. Explicitly, the two spherical distances from $\psi(u)$ to the antipodes sum to π and are at most cu and $\pi - cu$; both bounds are equalities. Writing $\psi(u) = \cos(cu)g + \sin(cu)n(u)$ with $n(u)$ a real orthogonal unit vector gives $\|\psi'\|^2 = c^2 + \sin^2(cu)\,\|n'\|^2$, so n is constant and fixed by the initial derivative. Consequently

$$W(u)g = (I - Q)g + \mathrm{e}^{-3iu}Qg. \tag{G2}$$

The real-sphere argument retains the Ramsey relative phase; projective Hilbert space would identify the two endpoints and would not give this conclusion.

Inserting Eq. (G2) into the interaction equation gives

$$Q\mathrm{e}^{-iuA}(I - Q)g = \mathrm{e}^{-3iu}(I - Q)\mathrm{e}^{-iuA}Qg. \tag{G3}$$

The two sides have disjoint supports and therefore both vanish for $0 < u < \tau$. Each initial half-line piece would stay confined to its original half-line.

Persistent confinement is excluded by semibounded time analyticity, the principle underlying the positive-observable alternative in Ref. [16]; we give its application directly here. For $f_- = (I - Q)g$ and any test vector k supported in $x > 0$, $F_k(z) = \langle k, \mathrm{e}^{-izA}f_-\rangle$ is bounded analytic for $\operatorname{Im} z < 0$, continuous on the real boundary, and zero on an open interval. Schwarz reflection across a smaller interval with zero boundary values followed by the identity theorem makes it identically zero. The heat evolution and all resolvents $(A + \lambda)^{-1}f_-$ therefore vanish on $x > 0$. Direct matching gives the cross-interface resolvent kernel

$$G_\lambda(x, y) = \frac{\mathrm{e}^{-qx}\mathrm{e}^{py}}{p + q}, \qquad x > 0 > y, \quad p = \sqrt{\lambda}, \quad q = \sqrt{\lambda + 1}. \tag{G4}$$

Thus $\int_{-\infty}^{0} \mathrm{e}^{py} f_-(y)\mathrm{d}y = 0$ for every $p > 0$. Analytic continuation to $p = a + i\xi$ and Fourier uniqueness for the exponentially damped half-line function prove $f_- = 0$. The reversed cross kernel gives the same conclusion for $f_+ = Qg$, initially for $q > 1$ and then by analytic continuation. This contradiction proves $\ker K = \{0\}$.

2. Smooth plateau and strictly interior profiles

For a smooth nondecreasing plateau profile replace $2Q - I$ by $2F - I$ in Eq. (G1). Its norm is at most one, so the curve from g to $-g$ has length at most π. Equality is necessary, and every speed equals c. It follows that

$$4\langle \mathrm{e}^{-iuB}g, F(1 - F)\mathrm{e}^{-iuB}g\rangle = 0, \qquad 0 \le u \le \tau. \tag{G5}$$

Let $a = \sup\{x : F(x) = 0\}$ and $b = \inf\{x : F(x) = 1\}$. Continuity and monotonicity give $a < b$ and $0 < F < 1$ on (a, b). Equation (G5) makes the evolution vanish there, including at zero time. The preceding time analyticity argument implies that $(B + \lambda)^{-1}g$ vanishes there for every $\lambda > 0$.

Let ϕ_- and ϕ_+ solve $(B + \lambda)\phi = 0$ and agree with $\mathrm{e}^{\sqrt{\lambda}(x-a)}$ for $x \le a$ and $\mathrm{e}^{-\sqrt{\lambda+4}(x-b)}$ for $x \ge b$, respectively. They are linearly independent: proportionality would give a negative-energy L^2 eigenfunction, contradicting $B \ge 0$. Their Wronskian is nonzero, and the Green formula, verified by its unit derivative jump, gives on (a, b)

$$[(B + \lambda)^{-1}g](x) = c_-(\lambda)\phi_+(x) + c_+(\lambda)\phi_-(x). \tag{G6}$$

Up to the same nonzero Wronskian factor the coefficients are

$$c_-(\lambda) = \int_{-\infty}^{a} \mathrm{e}^{\sqrt{\lambda}(y-a)}g(y)\mathrm{d}y,$$
$$c_+(\lambda) = \int_{b}^{\infty} \mathrm{e}^{-\sqrt{\lambda+4}(y-b)}g(y)\mathrm{d}y. \tag{G7}$$

There is no source integral inside (a, b) because g vanishes there. Linear independence forces both coefficients to vanish; the two Laplace uniqueness arguments prove $g = 0$. This proves injectivity for the fixed smooth plateau family.

For completeness, a multiplier satisfying $0 < F < 1$ almost everywhere, such as the finite analytic erf profile, gives an even shorter injectivity argument. The centered curve has speed at most c, but its initial speed is strictly less than c for every nonzero g, since $\int F(1 - F)|g|^2 > 0$. Continuity makes its total length strictly less than π, contradicting the antipodal endpoints. This observation supplies no quantitative finite-device later-round yield.

3. Complete thermal normalization after a fixed number of rounds

Let $V_\tau = (3/2)h_\tau(B_\infty)Ff_1$ and $w_m = K_\infty^{m-1}V_\tau$. The source is nonzero and injectivity gives $w_m \neq 0$ for every

fixed m. Strong convergence of the finite-box propagators, their uniform contraction bounds, and the source convergence give

$$K_{\tau,\ell}^{m-1} V_{0,\ell}(\tau) \longrightarrow w_m. \tag{G8}$$

The actual low columns after m rounds are

$$K_{\tau,\ell}^{m} u_E = b_E \mathrm{e}^{-i\tau E} K_{\tau,\ell}^{m-1} V_{E,\ell}(\tau). \tag{G9}$$

The scalar phase is $\mathrm{e}^{-i\tau E}$, not a replacement of K^m by a commuting polynomial. Multiplication by a contraction preserves the low Taylor approximation bounds and cannot increase the singular values of the high-sector column operator. The complete entropy bounds in Appendix E therefore apply after normalization by the positive source coefficient, giving

$$\begin{aligned} p_m &\sim \frac{\|w_m\|^2}{m_1^2}\frac{r}{s}, \qquad \operatorname{Tr}\sigma_m^2 \to 1, \\ S(\sigma_m) &= O_{m,F}(s^{-2}\log s + s^{5/2}\mathrm{e}^{-s\delta_F}) \to 0. \end{aligned} \tag{G10}$$

For the sharp wall one can take $\delta_F = 1/2$ and $m_1 = 1$. All fixed positive Rényi entropies also vanish. In particular $p_{m+1}/p_m \to \|w_{m+1}\|^2 / \|w_m\|^2 > 0$. There is no lower bound uniform in m, no limit with m growing, and no assertion at every higher resonance.

4. Two independent imperfect readouts

With $\nu_j \in [0,1]$ and $\epsilon_j = 1-\nu_j$, composition of the two maps in Eq. (20) is exactly

$$\begin{aligned} \mathcal{J}_{\nu_2}(\mathcal{J}_{\nu_1}(\rho)) &= \nu_1\nu_2 K^2\rho(K^*)^2 + \epsilon_1\nu_2 K\mathcal{D}(\rho)K^* \\ &\quad + \nu_1\epsilon_2\mathcal{D}(K\rho K^*) + \epsilon_1\epsilon_2\mathcal{D}^2(\rho). \end{aligned} \tag{G11}$$

All four terms are positive. If p_1, p_2 are the ideal probabilities, the third branch before its scalar coefficient has trace $p_1/2$ and the fourth has trace $1/4$. For the second branch use $U_B = U_A + 2K$ and $U_A u_E = \mathrm{e}^{-i\tau E}u_E$, giving $KU_B u_E = \mathrm{e}^{-i\tau E}Ku_E + 2K^2 u_E$. The U_A term alone gives its lower trace $p_1/4$; the inequality $\|a+2b\|^2 \le 2\|a\|^2 + 8\|b\|^2$ gives the upper trace $(3p_1+8p_2)/4$. Because both ideal probabilities are of order r/s, Eq. (21) makes the total false weight negligible relative to p_2, proving source retention and ideal leading yield. One effective visibility multiplying K^2 would be a different channel.

The normalized singly false branches have uniformly bounded entropy. For $K\mathcal{D}(\rho)K^*$ their low columns are combinations of Ku_E and K^2u_E, so two limiting source vectors suffice, and their complete tails are bounded by sums of the contracted analytic tails. Its trace is at least $p_1/4$, so normalization is controlled. The other singly false branch is a mixture of two unitary images of the one-round state. Their weights relative to the useful branch are $O(\epsilon_1+\epsilon_2)$, and their weighted entropies vanish.

The normalized all-false state $4\mathcal{D}^2(\rho)$ is a mixture of four unitary images of ρ. Concavity and the finite-label mixture bound give

$$S(\rho) \le S(4\mathcal{D}^2(\rho)) \le S(\rho) + \log 4. \tag{G12}$$

The partition and its first energy moment imply $S(\rho) = \log N + O(1)$, $N = 1/(r\sqrt{s})$. Consequently $\epsilon_1\epsilon_2 \log N = o(r/s)$, together with $\epsilon_1+\epsilon_2 \to 0$, pays the all-false entropy and the finite mixture-label entropy, proving sufficiency. Under Eq. (21), the all-false weight is asymptotic to $\epsilon_1\epsilon_2/(4p_2)$; entropy concavity and Eq. (G12) prove necessity of this strengthened product condition. These are results for the stated independent visibility channels, not correlated dephasing or unrecorded timing laws.

Appendix H: A validated smooth-wall and noise calculation

This calculation concerns the exact error-function wall in Eq. (39). It does not replace that wall by a finite-dimensional Hamiltonian. We describe the enclosures sufficiently to distinguish analytic error control from numerical convergence.

1. Resolving the transition with a controlled amplitude tail

Put $w = 1/40$, $e = \operatorname{erf}(5)$, $a_0 = 1-5w = 7/8$. Define a continuous plateau comparator by

$$F_0(y) = \begin{cases} 0, & y \le a_0, \\ [\operatorname{erf}((y-1)/w) + e]/(2e), & a_0 < y < 2-a_0, \\ 1, & y \ge 2-a_0. \end{cases} \tag{H1}$$

Monotonicity of the error function and direct subtraction inside the middle interval give

$$\|F_r - F_0\|_\infty \le \varepsilon = (1-e)/2 < 7.688\times 10^{-13}. \tag{H2}$$

This is an amplitude error. The mathematical validation collar has length $1/4$, whereas the physical 10–90 width is $2w\operatorname{erf}^{-1}(0.8) = 0.9061938024\ldots r$.

For the spectral solve use $H_a = -\partial_y^2 + aF_0$ with $a = 400, 1600$, inverse temperature $8/400$, and time $2\pi/1200$. Divide energies by 400 to return to Eq. (39). The middle interval is divided into 50 cells of length $h = 1/200$. In a cell let $x \in [0,1]$ and $z = -5 + j/5$, so the error-function argument is $z + x/5$. Taylor coefficients of $g(x) = \mathrm{e}^{-(z+x/5)^2}$ follow from $g' = -2(z+x/5)g/5$; integrating them supplies the coefficients of F_0. The initial coefficient is enclosed with Arb's error function. The

solution coefficients obey the exact recurrence

$$\begin{aligned}&(n+1)(n+2)u_{n+2}\\&= h^2\Big(a\sum_{j=0}^{n} F_j u_{n-j}\\&\quad -Eu_n\Big).\end{aligned}\tag{H3}$$

The value and derivative polynomials are retained through degree 48, with u_{49} computed for the latter.

For an explicit common remainder, take the complex radius $R = 4$ in the local variable. On that disk the error-function argument has imaginary part at most $b = 4/5$. Vertical integration from its real part gives

$$\begin{aligned}|\operatorname{erf}\zeta| &\le 1 + 2be^{b^2}/\sqrt{\pi},\\ |F_0| &\le M_F^* = \frac{e+1+2be^{b^2}/\sqrt{\pi}}{2e}.\end{aligned}\tag{H4}$$

For $Y = (u, u_x)$, the analytic first-order system has matrix infinity norm at most $M = \max\{1, h^2(|E| + aM_F^*)\}$. Gronwall along a radius and Cauchy's coefficient estimate imply the uniform error bound for both retained components

$$\eta \le \frac{e^{RM}\,\|Y(0)\|_\infty}{(R-1)R^{48}} \qquad (0 \le x \le 1).\tag{H5}$$

The physical derivative error is η/h.

To prevent interval wrapping, the propagated vector is recentered at each cell endpoint, carrying a scalar Euclidean error in $(u, u_y/k_*)$, with a fixed positive integer k_* for that eigenvalue bracket. The symmetric part of the real generator has largest eigenvalue at most

$$\frac{|k_*^2 - E| + a}{2k_*}.\tag{H6}$$

An incoming error is therefore multiplied by at most the exponential of h times this bound. Local endpoint ball radii and Eq. (H5) are added outwardly. The same growth bound controls every interior point used in integrals. Energy uncertainty is retained in each recurrence and exponent; recentering discards no error term.

2. Root labels, overlaps, and the full accepted Gram matrix

The left plateau solution is $\sin(\sqrt{E}\,y)$ and the right solution is boundary-adapted sine or hyperbolic sine. At the right collar endpoint the matching determinant is

$$\begin{cases} u_b\cos(qa_0) + v_b\sin(qa_0)/q, & E > a, \quad q = \sqrt{E-a},\\ u_b + v_b\tanh(\kappa a_0)/\kappa, & E < a, \quad \kappa = \sqrt{a-E}.\end{cases}\tag{H7}$$

The threshold is its continuous limit. Floating integration only proposes rational root brackets. The analytic ODE recomputes their endpoint signs. The free plateau nodes are counted exactly; on each collar cell, $u'' + (E - aF_0)u = 0$ and $E - aF_0 \le E$ imply that two zeros are separated by at least $\pi/\sqrt{E}$. The checker verifies that the cell is shorter than this distance and that its endpoint values are nonzero. Opposite signs then mean exactly one zero, equal signs none. Consecutive Sturm indices, ordered disjoint brackets, and the first omitted neighbor certify the spectral prefix without missing an intermediate level.

Cell overlap integrals multiply the enclosed polynomials. If their supremum bounds are U, V and errors η, ζ, the missing integral is at most $h(\eta V + \zeta U + \eta\zeta)$. Exact signed sine and hyperbolic-sine integrals supply both plateaus. This constructs normalization and signed overlaps for 30 initial and 126 final retained modes. The first omitted final eigenvalue and the overlap identity bound every omitted final overlap mass R_n by Eq. (I2), replacing Q by F_0. In the H_a energy units the numerator is $1200^2\,\|F_0u_n\|^2$; in dimensionless energies, after division by 400, it is $9\,\|F_0u_n\|^2$. The numerator norm is bounded by the full collar and right-plateau masses. Equation (I3) bounds every omitted initial Gibbs weight.

There is a useful factor of two in the Gram tail. For real eigenfunctions let $V_{nm}(t) = \sum_j O_{jn}O_{jm}e^{-it\lambda_j}$ in dimensionless energies. The exact initial-basis Gram matrix is

$$\begin{aligned}G_{nm} &= \langle K_t u_n, K_t u_m\rangle\\ &= \frac{\delta_{nm}}{2} - \frac{1}{4}\left[e^{itE_n}V_{nm} + \overline{e^{itE_m}V_{nm}}\right].\end{aligned}\tag{H8}$$

The missing part of V_{nm} is at most $\sqrt{R_nR_m}$, so the Gram error is at most $\sqrt{R_nR_m}/2$. Let $a_n = e^{-sE_n}$ and $T = \sum_n a_nG_{nn}$. Positivity gives

$$p = T/Z_1, \qquad \operatorname{Tr}\sigma^2 = \frac{\sum_{n,m} a_na_m|G_{nm}|^2}{T^2}.\tag{H9}$$

For a lower purity bound the retained (n, m) entry contributes $[\max\{0, |G_{nm}^{\text{prefix}}| - \sqrt{R_nR_m}/2\}]^2$, where the underline denotes its interval lower bound. All missing squared entries are nonnegative. Upper-bound T by the retained diagonal upper sum plus the full omitted initial trace, since $G_{nn} \le 1$. Lower and upper partition bounds give the acceptance interval. Input purity is $Z_1(2s)/Z_1(s)^2$, with both heat tails paid.

3. Passing to the actual analytic wall

Let $S = e^{-sA/2}/\sqrt{Z_1}$ and $S_0 = e^{-sA_0/2}/\sqrt{Z_{1,0}}$ be Hilbert–Schmidt purifications. Min–max and Eq. (H2) give $e^{-s\varepsilon}Z_{1,0} \le Z_1 \le e^{s\varepsilon}Z_{1,0}$. Duhamel at heat time $t = s/2$, with Schatten Hölder, bounds the integrand by

$$\begin{aligned}&\left\|e^{-(t-u)A}(F_r - F_0)e^{-uA_0}\right\|_2\\ &\quad\le \varepsilon Z_1^{(t-u)/(2t)}Z_{1,0}^{u/(2t)}\\ &\quad\le \varepsilon e^{(t-u)\varepsilon}\sqrt{Z_{1,0}}.\end{aligned}\tag{H10}$$

After integration and normalization of the two vectors this gives

$$\|S - S_0\|_2 \le d_{\rm prep} := s\varepsilon e^{s\varepsilon/2}. \tag{H11}$$

Unitary Duhamel also gives $\|K_\tau(F_r) - K_\tau(F_0)\| \le (5/2)|\tau|\varepsilon$. Thus the unnormalized output purifications differ by at most $h = d_{\rm prep} + (5/2)|\tau|\varepsilon$.

More generally, if a reference output purification has squared norm in $[p_-, p_+]$ and its state has purity at least P_-, a vector error at most $h < \sqrt{p_-}$ yields

$$(\sqrt{p_-} - h)^2 \le p \le (\sqrt{p_+} + h)^2, \qquad \operatorname{Tr}\sigma^2 \ge P_- - \frac{2h}{\sqrt{p_-}}. \tag{H12}$$

The probability bounds are the norm triangle inequalities. For the state, project the reference vector onto the line through the new one; the sine of their angle is at most $h/\sqrt{p_-}$. The pure-state trace-distance identity and partial-trace contractivity give the displayed purity loss, because purity is 1-Lipschitz in trace norm on density operators:

$$|\operatorname{Tr}(\sigma^2 - \rho^2)| = |\operatorname{Tr}[(\sigma-\rho)(\sigma+\rho-\mathbb{I})]| \le \|\sigma - \rho\|_1. \tag{H13}$$

The last inequality uses $-\mathbb{I} \le \sigma + \rho - \mathbb{I} \le \mathbb{I}$. Input purity changes by at most $2d_{\rm prep}$. These bounds give the actual analytic-wall nominal acceptance $0.0025635744408 < p_0 < 0.0025635744431$, nominal purity $P_0 > 0.9664657000$, and input purity below 0.386060.

4. Unrecorded time distributions and recorded outcomes

Since A commutes with its Gibbs square root, $B = A + 3F_r$ and unitarity give Eq. (41) by differentiating in Hilbert–Schmidt norm. In particular $\|(K_t - K_\tau)S\|_2 \le L|t - \tau|$. For any independent duration law μ supported on $[\tau-\Delta, \tau+\Delta]$, regard $K_t S$ as a vector in $L^2(\mu; \mathfrak{S}_2)$. Its distance from the constant reference $K_\tau S$ is at most $L\Delta$. Partial trace over this duration register gives exactly the unrecorded mixture $\overline{R} = \int K_t \rho K_t^* \mu(\mathrm{d}t)$. Equation (H12) therefore applies to this mixture, not just separately to states in duration-dependent frames.

Both moments in L are bounded for the actual F_r. For retained comparator energies E_n in dimensionless units, the actual unnormalized second-energy contribution is bounded by $(E_n + \varepsilon)^2 e^{-s(E_n - \varepsilon)}$. For omitted modes, put $c = sr^2\pi^2/4$, $x_0 = r^2\pi^2(n_0+1)^2/4$ and $q_0 = e^{-c(2n_0+3)}$. The checker verifies $x_0 > 2/s$. Monotonicity of $x^2 e^{-sx}$ above x_0 and $(n_0 + 1 + j)^4 \le (n_0+1)^4(j+1)^4$ give the full tail

$$x_0^2 e^{-c(n_0+1)^2} \frac{1 + 11q_0 + 11q_0^2 + q_0^3}{(1-q_0)^5}. \tag{H14}$$

Divide by the actual partition lower bound $e^{-s\varepsilon} Z_{1,0,-}$. For $\operatorname{Tr}\rho_0 F_0^2$, bound F_0^2 by its right endpoint in each validation cell and multiply by the certified cell norm. Add the entire right-plateau norm, where $F_0 = 1$, and the complete omitted Gibbs tail. Then

$$\operatorname{Tr}\rho F_r^2 \le \operatorname{Tr}\rho_0 F_0^2 + 2d_{\rm prep} + 2\varepsilon. \tag{H15}$$

The resulting bounds are $\operatorname{Tr}\rho A^2 < 0.015989357$, $\operatorname{Tr}\rho F_r^2 < 0.003950444$, and $L < 0.220727856$.

To include visibility and detector assignments, set

$$\overline{D} = \int \frac{U_A \rho U_A^* + U_B \rho U_B^*}{2} \mu(\mathrm{d}t), \qquad \operatorname{Tr}\overline{D} = 1. \tag{H16}$$

Both branches use the same duration law. With false-positive probability e and false-negative probability d, the recorded dark operator is exactly

$$R_{\rm obs} = a\overline{R} + b\overline{D}, \qquad a = (1-d-e)\nu, \quad b = (1-d-e)(1-\nu)/2 + e. \tag{H17}$$

This follows from $(1-d)\mathcal{J}_\nu + e\mathcal{J}_{\rm bright}$ and $\mathcal{J}_\nu + \mathcal{J}_{\rm bright} = \overline{D}$. For $d + e \le 1$ both coefficients are nonnegative. Positivity of the Hilbert–Schmidt cross term gives

$$\operatorname{Tr}\sigma_{\rm obs}^2 \ge \left(\frac{a\overline{p}}{a\overline{p} + b}\right)^2 \operatorname{Tr}\overline{\sigma}^2, \qquad p_{\rm obs} = a\overline{p} + b. \tag{H18}$$

Using $\Delta = 0.001$, $\nu = 0.9999$, $e = 10^{-5}$, $d = 0.01$ and outward endpoints in these monotone formulas gives Eq. (40). All probabilities are conditional on the specified physical channels; generic correlated preparation or detection errors are different models.

The accompanying scripts enclose every root, ODE remainder, overlap, Gram entry, Gibbs tail, and scalar propagation at 256-bit ball precision. Saved source hashes identify the scripts actually replayed. Their finite prefixes are computational devices inside the inequalities above and never project the physical output.

Appendix I: Finite spectral enclosures and statistical reconstruction

This appendix distinguishes deterministic continuum error from statistical counting error. Both are needed to use the source reconstruction in an actual finite experiment. Throughout this appendix the device is the sharp-wall Hamiltonian in Eq. (1); the noisy smooth example is separate.

1. Complete finite-device probabilities

For each initial eigenstate, completeness and $B - A = 3Q$ give

$$\sum_j |O_{jn}|^2 = 1, \qquad (\lambda_j - E_n) O_{jn} = 3\langle \phi_j, Q u_n\rangle, \qquad \sum_j (\lambda_j - E_n)^2 |O_{jn}|^2 = 9\,\|Q u_n\|^2, \tag{I1}$$

$$O_{jn} = \langle \phi_j, u_n\rangle.$$

Take a complete initial prefix of length n_0 and a complete final prefix of length j_0, including certified lower endpoints for the first omitted levels. If $\lambda_{j_0+1} \geq L_* > E_n$, the missing final overlap mass obeys

$$R_n := \sum_{j>j_0} |O_{jn}|^2 \leq \min\left\{1 - \sum_{j\leq j_0} |O_{jn}|^2, \frac{9\,\|Qu_n\|^2}{(L_* - E_n)^2}\right\}. \tag{I2}$$

The upper interval endpoint of each expression is used. For the full omitted initial sector, min–max comparison with the free Dirichlet interval gives

$$T_{n_0} := \sum_{n>n_0} \mathrm{e}^{-sE_n} \leq \frac{\mathrm{e}^{-b(n_0+1)^2}}{1-\mathrm{e}^{-b(2n_0+3)}}, \qquad b = sr^2\pi^2/4. \tag{I3}$$

Indeed the successive exponents beyond n_0+1 increase by at least $b(2n_0+3)$. This tail includes every omitted state above the barrier; retained states in that sector remain in the finite sum.

Let Z_- and Z_+ enclose $\sum_{n\leq n_0} \mathrm{e}^{-sE_n} + [0, T_{n_0}]$, and define the finite positive numerator

$$P_{n_0 j_0}(t) = \sum_{n\leq n_0, j\leq j_0} \mathrm{e}^{-sE_n} |O_{jn}|^2 \sin^2[t(\lambda_j - E_n)/2]. \tag{I4}$$

Equations (28) and (I2) then give

$$\frac{P_{n_0 j_0}(t)}{Z_+} \leq p_{r,s}(t) \leq \frac{P_{n_0 j_0}(t) + \sum_{n\leq n_0} \mathrm{e}^{-sE_n} R_n + T_{n_0}}{Z_-}. \tag{I5}$$

The finite sums themselves are evaluated with outward interval arithmetic, so their lower and upper endpoints are used on the corresponding sides. Neither tail uses oscillatory cancellation.

For clarity, all eigenvalues can be enclosed without a spatial discretization. For a wall of height h on the right half of $(0,2)$, write $k = \sqrt{E}/r$, $q = \sqrt{E-h}/r$. Matching $\sin(ky)/k$ with the right boundary-adapted function $\sin[q(2-y)]/q$ gives the entirely continued determinant

$$\cos k\,\frac{\sin q}{q} + \frac{\sin k}{k}\cos q = 0. \tag{I6}$$

Imaginary q gives the hyperbolic formula; both sine quotients are continued with value one at zero. Rational sign brackets, disjoint in increasing order, and certified Sturm node counts prove that the enclosed roots have consecutive indices starting at one. Analytic sine and hyperbolic-sine integrals give their normalization and signed cross overlaps. Floating-point root searches only propose brackets; signs and indices are rechecked with ball arithmetic [18].

The data in Sec. VII use $n_0 = 64$ and $j_0 = 1260$ at $(r,s) = (1/200, 128)$. The weighted, scaled Gibbs-tail allowance is below 5.612×10^{-11}; the final spectral-tail allowance is below 1.297776×10^{-5}. These errors are already included in Eq. (I5), not appended a second time.

2. The exact source quadrature and its tails

For $c = 1$, use $T = 32$, the 256 nodes $t_i = (2i+1)/16$, $i = 0, \ldots, 255$, and the positive weights in Eq. (36). Their exact sum is

$$W = \frac{4}{\pi}\left[\mathrm{Si}(32) - \frac{1-\cos 32}{32}\right], \qquad W = 1.95959365427766\ldots. \tag{I7}$$

The weight is fixed by the reconstruction identity, not fitted to the value of $\Phi(1)$.

For the low-energy part of $C(t)$, substitute $E = 4\sin^2\theta$ in Eq. (12). Its integrand on $[0, \pi/2]$ becomes

$$\frac{9t^2}{\pi}\,\frac{\sin^2\theta\cos\theta}{(1+2\cos\theta)^2}\,\mathrm{sinc}^2(2t\sin^2\theta), \tag{I8}$$

where $\mathrm{sinc}\, z = \sin z/z$ is continued at zero. Above four, set $E = 4+q^2$, $k = \sqrt{4+q^2}$. The measure density in q is

$$h(q) = \frac{18q(1+kq)}{\pi(q^2+4)^2(k+q)(q^2+1)^2}. \tag{I9}$$

Integrate the actual sine-square response over $0 \leq q \leq 4$. The complete omitted mass is enclosed by the nonsingular finite integral, obtained from $q = 1/u$,

$$\nu((20,\infty)) = \int_0^{1/4} \frac{18u^4(u^2+\sqrt{1+4u^2})}{\pi(1+4u^2)^2(1+\sqrt{1+4u^2})(1+u^2)^2}\,\mathrm{d}u. \tag{I10}$$

It lies within $0.0004026117 \pm 4.36 \times 10^{-11}$. For every time node its missing sine-square contribution belongs to $[0, \nu((20,\infty))]$.

For $\Phi(1)$ the low-angle integral is split at $\theta = \pi/6$. Before and after this point the integrands are respectively

$$\frac{9\cos\theta}{\pi(1+2\cos\theta)^2}, \qquad \frac{9\cos\theta}{4\pi\sin^2\theta(1+2\cos\theta)^2}. \tag{I11}$$

The complete mass above four from Eqs. (I9) and (I10) is then added. Analytic ball quadrature [19] encloses all finite integrals, producing Eq. (25) without a missing high-energy tail.

Writing $S_C = \sum_i w_i C(t_i)$, the certificate keeps separate enclosures for $S_C - \Phi(1)$, $S - S_C$, and $\sum_i w_i|\lambda p(t_i) - C(t_i)|$. At the stated device,

$$\begin{aligned} |S - \Phi(1)| &< 0.000479010749, \\ |S_C - \Phi(1)| &< 0.047699256289, \\ \sum_i w_i|\lambda p(t_i) - C(t_i)| &< 0.047233223348. \end{aligned} \tag{I12}$$

The small direct error includes cancellation between two biases of opposite sign. The separate absolute budget is below 0.094932479636. Because both the full source target and the finite source grid are enclosed, this already pays the source time-window and mesh errors; a further time-tail allowance would double-count them. These numerical statements concern this known sharp source, not a uniformly stable inverse of an unspecified potential.

3. Exact count intervals and the preparation budget

Assume independent trials with the same node probabilities $p(t_i)$. Let X be the dark-event indicator after randomly choosing a node with probability w_i/W. Since this choice is made independently on every trial, X is Bernoulli with $\bar{p} = \sum_i (w_i/W)p(t_i)$; a mixture of Bernoulli indicators is still Bernoulli. Therefore $k = \sum_{j=1}^{n} X_j$ is binomial. One exact equal-tailed interval [20] is

$$L_n(k) = \begin{cases} 0, & k = 0, \\ \mathrm{BetaQuantile}(\alpha/2; k, n-k+1), & k > 0, \end{cases}$$
$$U_n(k) = \begin{cases} 1, & k = n, \\ \mathrm{BetaQuantile}(1-\alpha/2; k+1, n-k), & k < n. \end{cases} \tag{I13}$$

Each endpoint inverts one binomial tail of probability $\alpha/2$; the union bound gives coverage at least $1-\alpha$. In particular $U_n(0) = 1 - (\alpha/2)^{1/n}$, so zero successes do not require a Gaussian approximation. Combining the interval with deterministic $d_\pm$ proves Eq. (37).

For the predeclared budget, a centered Bernoulli variable Z has $\mathbb{E}|Z|^j \le \mathrm{Var}\, Z$ for every integer $j \ge 2$. Expanding the exponential and using $j! \ge 2\,3^{j-2}$ gives

$$\log \mathbb{E}\mathrm{e}^{uZ} \le \frac{u^2 \,\mathrm{Var}\, Z}{2(1-|u|/3)}, \qquad |u| < 3. \tag{I14}$$

Exponential Markov for the two signs and independence then yield

$$\mathbb{P}(|\widehat{S} - S| \ge \epsilon) \le 2\exp\left[-\frac{n\epsilon^2}{2\lambda W(D+\epsilon/3)}\right], \tag{I15}$$

because $\lambda W \bar{p} = S \le D$. This proves Eq. (38). A positive upper bound on the *actual* scaled response is required: subtracting a known background mean would not remove its Bernoulli variance.

The implementation uses exact dyadic weights $\widetilde{w}_i = 2^{-80}\lfloor 2^{80} w_i \rfloor$, with each integer numerator certified uniquely. If $p_i \le U_i/\lambda$ and $|w_i - \widetilde{w}_i| \le e_i$, the added bias is at most

$$B_w = \sum_i e_i U_i. \tag{I16}$$

Run the interval and budget formulas with $\widetilde{W}$ and the exact dyadic sampling law, widening the bias endpoints by B_w. Here $B_w < 5.35 \times 10^{-23}$. A sampling-law implementation with total-variation error ζ would require the further allowance $\lambda\widetilde{W}\zeta$; the stated budgets specify the exact discrete law.

For $\alpha = 0.05$, use $\epsilon = 0.05$ with the first line of Eq. (I12), or $\epsilon = 0.005$ with its two separate absolute errors. Outward rounding of Eq. (38) gives the counts displayed in Sec. VII, and respective total error bounds 0.050479011 and 0.099932480. No measured count is fabricated by this design calculation.

4. Allocation and duration are part of the resource model

The pooled protocol is simple but not universally variance optimal. If $v_i = p_i(1-p_i)$ and fixed counts $n_i = nq_i$ are used at the nodes, the idealized unrounded variance is

$$\frac{\lambda^2}{n}\sum_i \frac{w_i^2 v_i}{q_i}. \tag{I17}$$

Cauchy–Schwarz minimizes it at $q_i \propto w_i\sqrt{v_i}$ on nonzero-variance nodes. This is an oracle allocation; learning the p_i requires its own data. Equal shots have between one and m times that minimal variance for m nodes, provided at least one $v_i > 0$. Randomizing with $q_i = w_i/W$ adds, relative to the same fixed allocation, the nonnegative variance $\lambda^2 W^2 \,\mathrm{Var}_q(p_i)/n$.

A usable confidence-bound design uses certified U_i instead. For any positive distribution q, one trial has value $Y = \lambda w_I X/q_I$, expectation S, variance at most $\lambda A(q)$, and range at most $\lambda B(q)$, where

$$A(q) = \sum_i w_i^2 U_i/q_i, \qquad B(q) = \max_i w_i/q_i. \tag{I18}$$

The same argument as Eq. (I15) replaces $W(D+\epsilon/3)$ by $A(q) + \epsilon B(q)/3$. Its minimization is a finite convex problem. If all $U_i = 0$, then $S = 0$ and no sampling is required. Otherwise, at fixed $b \ge W$ and constraints $q_i \ge w_i/b$, the minimizer has

$$q_i(b) = \max\{w_i/b,\ w_i\sqrt{U_i}/t_b\}, \qquad \sum_i q_i(b) = 1, \tag{I19}$$

with the limiting convention when a set of U_i vanishes. Minimizing $A(q(b)) + \epsilon b/3$ over $b \ge W$ gives the best bound of this form, not an optimal quantum sensing theorem. The displayed numerical budgets use the simpler pooled design.

Each trial consumes one complete Gibbs preparation, a fresh pure ancilla, a controlled pair of branches, and a readout. At the 256-node design the mean dimensionless duration is 2.616668820164279…, and its maximum is 31.9375. For any fixed n, the mean total controlled duration is

$$\mathbb{E}\mathcal{T} = n\frac{\hbar r^2}{E_0}\sum_i q_i t_i, \qquad \mathcal{T} \le n\frac{\hbar r^2 T}{E_0}. \tag{I20}$$

Preparation, reset, switching, and readout overheads are additional. The probability of retaining a state is not a license to omit failed trials from either the confidence interval or this cost.

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